Supplementary Material

Quantification corrections did not meaningfully reduce time-budget error of reused accelerometer classifiers in cats and dogs

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Conventions unless stated otherwise: AE = mean absolute error of class proportions; d = AE_reference − AE_arm (positive: the arm is better).

Decisions: better / worse = Holm-adjusted $p < 0.05$ and $|d| \geq 0.01$ in that direction; equivalent = 95% interval within $[-0.01, 0.01]$; otherwise inconclusive.

Scenarios: S0 random seconds (leaky); S1 new individual; S2 harness → collar; S3 Smit → Dunford; S4 dog → cat (open: target maintenance kept; closed: removed); R-D within Dunford; S1-closed: S1 without maintenance; L2: seven Smit classes; K-L1, K-L2: two and five dog classes. [exploratory] marks results based on the Dunford data.

Sample types: natural 30-min and 2-min windows, cat samples (Dunford), dog recordings, and artificial-prevalence (APP) samples of 120 s.

Values are rounded for print. The same tables at full numerical precision are in Supplementary_Material.xlsx and in the code repository (`https://github.com/masoudmaleki/cat-dog-time-budget`).

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Table S1. D17 versus the axis-mapped 32 features in harness-to-collar transfer (S2, Smit, three classes, 68 natural 30-min windows).

d = AE_mapped32 − AE_D17 (mean over individuals; positive: D17 better). Holm adjustment over the three estimators.

estimator	n windows	AE D17	AE mapped32	n individuals	d	ci low	ci high	p	p min attainable	p holm	family size	category
CC	68	0.0405	0.0683	12	+0.025	−0.002	+0.067	0.214	4.9×10^{-4}	0.393	3	inconclusive
ACC	68	0.0528	0.0904	12	+0.032	−0.002	+0.088	0.196	4.9×10^{-4}	0.393	3	inconclusive
PACC	68	0.1155	0.2016	12	+0.086	+0.057	+0.118	4.9×10^{-4}	4.9×10^{-4}	0.001	3	better

Table S2. Analysis history: every hypothesis and criterion tested, when it was specified, its result and where it is reported.

R02–R08 are successive analysis rounds. 'after Rxx results were seen' marks analyses defined after earlier results were available. 'this table only': the result is reported nowhere else. Effects are differences in AE with 95% individual-bootstrap intervals.

round	hypothesis or criterion	definition	when specified	result	location in paper
R02	H1: in S1-collar (seven classes), do ACC, PACC or EMQ have lower AE than CC?	Smit L2, 32 features, 68 windows	pre-specified (task text)	No: CC − ACC −0.0173 [−0.0266; −0.0081] (CC better); PACC and EMQ also worse than CC	main text
R02	H2: is the advantage of ACC smaller in S2 than in S1?	same	pre-specified	S2 − S1 ACC difference +0.0004 [−0.0163; 0.0157]: no difference shown	this table only
R02	H3: agreement between macro-F1 and AE rankings (Spearman)	cat level	pre-specified	see r02_h3_spearman.csv in the code repository	code repository
R03	A: code check (ACC with the confusion matrix of S0 itself)	sanity check	after R02 results were seen	pooled ACC AE 1.4e-12 (no error); in single-cat windows CC < ACC even in S0	this table only
R03	B: does an oracle confusion matrix (from target labels) beat CC?	diagnostic of separability	after R02 results were seen	No: S1-collar CC − oracle +0.0059 [−0.0172; 0.0281]. Superseded: this oracle leaked test data and was replaced by the cross-fitted oracle in R04	not reported (superseded)
R03	C: does calibration (isotonic, temperature) remove the problem?	ECE, Brier score	after R02 results were seen	No: no calibrated estimator beat CC	this table only
R03	D/H4: target label budget (T-ACC; retraining)	m = 60–1800 s	after R02 results were seen	only retraining gained, in S1-collar (m = 1800: +0.0333 [0.0170; 0.0535]); T-ACC did not	main text (superseded by R05–R07 designs)
R03	E: S2 axis mapping and invariant features	S2 mapped; S2 invariant	after R02 results were seen	mapping did not improve AE (raw − mapped −0.0191 [−0.0392; 0.0027])	this table only
R04	Does ACC with a cross-fitted oracle matrix (ACC-xfit) beat CC?	bias–variance decomposition	after R03 results were seen	No in S1; only in S2-mapped +0.0571 [0.0145; 0.1059]	supplement (Table S11)
R04	Breakdown criterion (bias^2 of CC > variance of ACC)	analytic criterion	after R03 results were seen	hit rate 0.485 (baseline 0.663): weak	not reported
R05	Gate A (feature port)	median VM_Mean in inactive seconds	pre-specified	passed (Smit 0.978; Dunford 0.964; dog 1.012 g)	Methods
R05	H5: do source-calibrated arms beat CC? (prediction: no)	S3, S4	pre-specified (prediction written)	only S3-ACC +0.0292 passed; inconclusive under the decision rule adopted in R06	main text (decision-rule categories)
R05	H6: smallest m at which the best study-level correction is significant	S3, S4	after R04 results were seen	none at any m	this table only
R05	H7: individual-level calibration better than study-level calibration (prediction: yes)	S4-Smit	after R04 results were seen	No (prediction not met). Later judged an invalid test design	not reported (invalid design)
R05	H8: retraining is the best arm	S1, S2, S3, S4	after R04 results were seen	yes in S1, S2 and S4-Smit; no or inconclusive in S3 and Dunford (reclassified in R06 F)	main text
R05	H9: does the artificial-prevalence protocol favour corrections?	Smit S1 bridge	after R04 results were seen	Yes: APP +0.0601 [0.0413; 0.0765]; natural 2-min −0.0176 [−0.0447; 0.0002]	main text
R06	A1 gate (explaining 0.0260 vs 0.0290)	count check	after R05 results were seen	passed (an analysis error in R05 was found and corrected)	this table only
R06	A2: cost of D17 (S2, D17 vs axis-mapped 32 features)	68 windows	after R05 results were seen	PACC +0.0864 better with D17; CC and ACC inconclusive	supplement (Table S1)
R06	B: duplicated blocks in the Dunford file (v1 → v2)	data quality	after R05 results were seen	all 413 over-full seconds contain duplicated rows; v2 used	main text (Methods, Results)
R06	C: does the naive baseline (N-target) beat CC? (each scenario tested alone)	CC vs N-target	after R05 results were seen	CC better in S1, S2, S4-Smit-closed and R-D; S3 and Dunford transfer inconclusive. In Table 3, N-target is tested in the family of the source-calibrated arms (except the seven-class collar row)	main text
R06	D: gain versus shift (δ, κ, τ rule)	Spearman correlation; frozen τ	after R05 results were seen	the rule did not beat CC in the held-out scenarios	supplement (Table S9)
R06	E1: weighted retraining	m = 300, 1800 s	after R05 results were seen	S4-Smit-open, individual level, m = 1800: +0.0142 better; others inconclusive or equivalent	supplement (Table S3)
R06	E2: target-only model (paired) − CC	m = 300, 1800 s	after R05 results were seen	S4-Smit-open, individual level, m = 1800: +0.0547 better; S4-Smit-closed worse	main text

continued

round	hypothesis or criterion	definition	when specified	result	location in paper
R06	E3: class coverage ($\geq$ 20 s of every class)	m = 1800 s, S1	after R05 results were seen	2.1% of label sets contain $\geq$ 20 s of every class	main text
R06	E4: S1-closed − S4-Smit-closed (cost of the species shift)	68 windows	after R05 results were seen	CC equivalent (−0.0024 [−0.0073; 0.0041])	main text (Results, dog- vs cat-trained models)
R06	F: reclassification of R05 H5 and H8 under the decision rule	decision rule	after R05 results were seen	the S3-ACC gain no longer held (inconclusive)	main text
R07	§1 target-only − CC and − weighted retraining (S3, S4)	m = 300, 1800 s	after R06 results were seen	see Table 5 and Results	main text
R07	§3 Dunford split unit and duplicated blocks	random 60/40 split vs leave-one-cat-out	after R06 results were seen	see Table S10	Discussion; Table S10
R07	§4 robustness with gradient boosting and logistic regression	source-calibrated arms	after R06 results were seen (interpretation pre-specified)	see Tables S3 and S4	main text; Tables S3 and S4
R07	§5 replication of H5 and H9 in dogs (n = 44)	K-L2, K-L1, S2-dog	after R06 results were seen (interpretation pre-specified)	see Tables 3 and 6 and Table S3	main text; Table S3
R07	§6 stratified target labelling	T-ACC, PACC-target and target-only, stratified	after R06 results were seen (interpretation pre-specified)	see Table 5	main text
R08	DM − CC (distribution matching, multiclass HDy with 10 bins)	12 primary families; decision rule; individual bootstrap and Holm over 3 tests	pre-specified (task text written before results)	better 1/12: S1 APP 120-s +0.0477; worse 3/12: S1 30-min −0.0325, S2 −0.0984, K-L2 recording −0.0289; equivalent 2/12: S4-Smit-closed −0.0066, K-L1 recording −0.0047; inconclusive 6/12: S1 2-min −0.0276, S1-L2 30-min −0.0130, K-L1 2-min +0.0089, K-L2 2-min +0.0069, S3 [exploratory] +0.0123, S4-Dunford-closed [exploratory] −0.0232	main text; Tables S3 and S5
R08	DR-APP − CC (descriptor: 9 deciles per feature; APP training samples)	12 primary families; decision rule	pre-specified (task text)	better 1/12: S1 APP 120-s +0.0341; worse 8/12: S1 2-min −0.2321, S1 30-min −0.1269, S2 −0.1862, S4-Smit-closed −0.0792, K-L1 2-min −0.1133, K-L1 recording −0.0664, K-L2 2-min −0.1070, K-L2 recording −0.0452; inconclusive 3/12: S1-L2 30-min −0.0287, S3 [exploratory] −0.1420, S4-Dunford-closed [exploratory] −0.1121	main text; Tables S3 and S6
R08	DR-natural − CC (descriptor: 9 deciles per feature; natural training windows)	12 primary families; decision rule	pre-specified (task text)	worse 9/12: S1 2-min −0.0207, S1 30-min −0.0507, S1 APP 120-s −0.1067, S2 −0.0409, S4-Smit-closed −0.0243, K-L1 2-min −0.0721, K-L1 recording −0.1225, K-L2 2-min −0.0542, K-L2 recording −0.0583; inconclusive 3/12: S1-L2 30-min −0.0020, S3 [exploratory] −0.0938, S4-Dunford-closed [exploratory] −0.1396	main text; Tables S3 and S6
R08	DR-h (secondary descriptor: per-class posterior histograms), APP and natural training samples	secondary family, 2 tests per cell	pre-specified (task text)	DR-APP-h: better 1/12: S1 APP 120-s +0.0478; worse 7/12: S1 2-min −0.0498, S1 30-min −0.0442, S2 −0.1070, S4-Smit-closed −0.0228, K-L1 2-min −0.0385, K-L1 recording −0.0321, K-L2 recording −0.0217; inconclusive 4/12: S1-L2 30-min −0.0338, K-L2 2-min −0.0219, S3 [exploratory] −0.0077, S4-Dunford-closed [exploratory] −0.0515. DR-natural-h: worse 6/12: S1 2-min −0.0200, S1 30-min −0.0435, S4-Smit-closed −0.0159, K-L1 2-min −0.0171, K-L1 recording −0.0699, K-L2 recording −0.0320; inconclusive 6/12: S1 APP 120-s −0.0140, S1-L2 30-min −0.0087, S2 −0.0540, K-L2 2-min −0.0044, S3 [exploratory] +0.0270, S4-Dunford-closed [exploratory] −0.0510	supplement (Table S3)
R08	Positive-contribution criterion for DR (a: better in Smit S1 30-min; b: better in dog K-L1 or K-L2 recordings; c: no 'worse' in any primary family; all for the same variant)	operational definition fixed in the task text	pre-specified (fixed before results were seen)	DR-APP: a no, b no, c no; DR-natural: a no, b no, c no; criterion not met	main text (Methods)
R08	Rule for intermediate outcomes (better in natural families / only under APP / only with the secondary descriptor)	choice of interpretation	pre-specified (task text)	applicable case: only under APP; natural cells better: 0; APP-only cells better: 2; secondary natural cells better: 0	main text

Table S3. All statistical comparisons of an estimator with a reference arm: effects, 95% intervals, Holm-adjusted p-values and decisions.

d = AE_reference − AE_arm, mean over individuals of per-individual differences (positive: the arm is better than the reference); 95% individual-bootstrap interval; exact sign-flip test, or 200,000 random sign assignments where stated; Holm adjustment within the family whose size is given. AE_CC_samples is the error of CC as a mean over samples, as in Tables 3 and 6 of the paper; AE_reference and AE_arm are on the basis given in AE_basis. Groups: A, Cells of Tables 3 and 6 (random forest); B, Dogs: other label and feature sets, and back harness to collar (S2-dog); C, Gradient boosting (HGB) and logistic regression (LR); D, Direct regression with the secondary descriptor (posterior histograms, DR-h); E, Retraining with labelled target seconds, unweighted and weighted; F, Labelling strategy: stratified minus random, each relative to its own CC on the same seconds. In group F, d = [AE_CC − AE_arm]stratified − [AE_CC − AE_arm]random, each term computed on the seconds that remained after that strategy's labelled seconds were removed (positive: stratified labelling gains more over its own CC). In the print version, * after the family size marks Monte Carlo p-values.

Grp	Clf	Scenario	Labels, features	Samples	Level, m	Arm	Reference	n	AE ref.	AE arm	d [95% CI]	p_{Holm} (family)	Decision
A	RF	S0 (leaky)		Smit 30-min (68)		PCC	CC	12	0.0211	0.0308	−0.010 [−0.014, −0.005]	0.008 (4)	inconclusive
A	RF	S0 (leaky)		Smit 30-min (68)		ACC	CC	12	0.0211	0.0256	−0.003 [−0.009, +0.002]	0.349 (4)	equivalent
A	RF	S0 (leaky)		Smit 30-min (68)		PACC	CC	12	0.0211	0.0329	−0.012 [−0.018, −0.004]	0.034 (4)	worse
A	RF	S0 (leaky)		Smit 30-min (68)		EMQ	CC	12	0.0211	0.0415	−0.021 [−0.036, −0.006]	0.034 (4)	worse
A	RF	S1		Smit 30-min (68)		PCC	CC	12	0.0260	0.0425	−0.015 [−0.025, −0.007]	0.004 (5)	worse
A	RF	S1		Smit 30-min (68)		ACC	CC	12	0.0260	0.0324	−0.006 [−0.012, −0.000]	0.064 (5)	inconclusive
A	RF	S1		Smit 30-min (68)		PACC	CC	12	0.0260	0.0527	−0.024 [−0.049, −0.007]	0.021 (5)	worse
A	RF	S1		Smit 30-min (68)		EMQ	CC	12	0.0260	0.0770	−0.044 [−0.110, −0.005]	0.044 (5)	worse
A	RF	S1		Smit 30-min (68)		N-target	CC	12	0.0260	0.1091	−0.083 [−0.101, −0.066]	0.002 (5)	worse
A	RF	S1		Smit 30-min (68)		DM	CC	12	0.0260	0.0649	−0.033 [−0.085, −0.003]	0.027 (3)	worse
A	RF	S1		Smit 30-min (68)		DR-APP	CC	12	0.0260	0.1505	−0.127 [−0.162, −0.098]	0.001 (3)	worse
A	RF	S1		Smit 30-min (68)		DR-natural	CC	12	0.0260	0.0807	−0.051 [−0.070, −0.031]	0.002 (3)	worse
A	RF	S2		Smit 30-min (68)		PCC	CC	12	0.0405	0.0745	−0.031 [−0.051, −0.014]	0.002 (5)	worse
A	RF	S2		Smit 30-min (68)		ACC	CC	12	0.0405	0.0528	−0.012 [−0.019, −0.004]	0.017 (5)	worse
A	RF	S2		Smit 30-min (68)		PACC	CC	12	0.0405	0.1155	−0.069 [−0.124, −0.025]	0.009 (5)	worse
A	RF	S2		Smit 30-min (68)		EMQ	CC	12	0.0405	0.1868	−0.134 [−0.232, −0.047]	0.009 (5)	worse
A	RF	S2		Smit 30-min (68)		N-target	CC	12	0.0405	0.1091	−0.067 [−0.084, −0.051]	0.002 (5)	worse
A	RF	S2		Smit 30-min (68)		DM	CC	12	0.0405	0.1480	−0.098 [−0.186, −0.030]	0.007 (3)	worse
A	RF	S2		Smit 30-min (68)		DR-APP	CC	12	0.0405	0.2313	−0.186 [−0.221, −0.157]	0.001 (3)	worse
A	RF	S2		Smit 30-min (68)		DR-natural	CC	12	0.0405	0.0808	−0.041 [−0.061, −0.021]	0.007 (3)	worse
A	RF	S3 [exploratory]		Dunford cat sample		PCC	CC	8	0.1442	0.1758	−0.032 [−0.073, +0.003]	0.719 (5)	inconclusive
A	RF	S3 [exploratory]		Dunford cat sample		ACC	CC	8	0.1442	0.1056	+0.039 [+0.003, +0.081]	0.352 (5)	inconclusive
A	RF	S3 [exploratory]		Dunford cat sample		PACC	CC	8	0.1442	0.1306	+0.014 [−0.025, +0.062]	0.734 (5)	inconclusive
A	RF	S3 [exploratory]		Dunford cat sample		EMQ	CC	8	0.1442	0.1924	−0.048 [−0.141, +0.049]	0.734 (5)	inconclusive
A	RF	S3 [exploratory]		Dunford cat sample		N-target	CC	8	0.1442	0.2274	−0.083 [−0.201, +0.033]	0.719 (5)	inconclusive
A	RF	S3 [exploratory]		Dunford cat sample		DM	CC	8	0.1442	0.1319	+0.012 [−0.030, +0.059]	0.680 (3)	inconclusive
A	RF	S3 [exploratory]		Dunford cat sample		DR-APP	CC	8	0.1442	0.2862	−0.142 [−0.282, +0.008]	0.375 (3)	inconclusive
A	RF	S3 [exploratory]		Dunford cat sample		DR-natural	CC	8	0.1442	0.2379	−0.094 [−0.220, −0.001]	0.375 (3)	inconclusive
A	RF	S4-Smit-open		Smit 30-min (68)		PCC	CC	12	0.0989	0.0991	−0.000 [−0.000, −0.000]	0.336 (5)	equivalent
A	RF	S4-Smit-open		Smit 30-min (68)		ACC	CC	12	0.0989	0.1001	−0.001 [−0.002, +0.000]	0.409 (5)	equivalent
A	RF	S4-Smit-open		Smit 30-min (68)		PACC	CC	12	0.0989	0.1002	−0.001 [−0.003, +0.000]	0.409 (5)	equivalent
A	RF	S4-Smit-open		Smit 30-min (68)		EMQ	CC	12	0.0989	0.1058	−0.006 [−0.008, −0.002]	0.039 (5)	equivalent
A	RF	S4-Smit-open		Smit 30-min (68)		N-target	CC	12	0.0989	0.1091	−0.021 [−0.052, +0.004]	0.409 (5)	inconclusive
A	RF	S4-Smit-closed		Smit 30-min (68)		PCC	CC	12	0.0089	0.0100	−0.001 [−0.002, +0.000]	0.844 (5)	equivalent
A	RF	S4-Smit-closed		Smit 30-min (68)		ACC	CC	12	0.0089	0.0105	−0.001 [−0.004, +0.002]	0.844 (5)	equivalent
A	RF	S4-Smit-closed		Smit 30-min (68)		PACC	CC	12	0.0089	0.0111	−0.002 [−0.005, +0.002]	0.844 (5)	equivalent
A	RF	S4-Smit-closed		Smit 30-min (68)		EMQ	CC	12	0.0089	0.0208	−0.010 [−0.014, −0.005]	0.023 (5)	worse
A	RF	S4-Smit-closed		Smit 30-min (68)		N-target	CC	12	0.0089	0.0410	−0.033 [−0.040, −0.027]	0.002 (5)	worse
A	RF	S4-Smit-closed		Smit 30-min (68)		DM	CC	12	0.0089	0.0166	−0.007 [−0.010, −0.003]	0.022 (3)	equivalent
A	RF	S4-Smit-closed		Smit 30-min (68)		DR-APP	CC	12	0.0089	0.0849	−0.079 [−0.110, −0.060]	0.001 (3)	worse

continued

Grp	Clf	Scenario	Labels, features	Samples	Level, m	Arm	Reference	n	AE ref.	AE arm	d [95% CI]	p_{Holm} (family)	Decision
A	RF	S4-Smit-closed		Smit 30-min (68)		DR-natural	CC	12	0.0089	0.0376	−0.024 [−0.042, −0.007]	0.022 (3)	worse
A	RF	S4-Dunford-open [exploratory]		Dunford cat sample		PCC	CC	8	0.1039	0.1078	−0.004 [−0.009, +0.000]	0.305 (5)	equivalent
A	RF	S4-Dunford-open [exploratory]		Dunford cat sample		ACC	CC	8	0.1039	0.1110	−0.007 [−0.011, −0.004]	0.078 (5)	inconclusive
A	RF	S4-Dunford-open [exploratory]		Dunford cat sample		PACC	CC	8	0.1039	0.1142	−0.010 [−0.014, −0.006]	0.078 (5)	inconclusive
A	RF	S4-Dunford-open [exploratory]		Dunford cat sample		EMQ	CC	8	0.1039	0.1295	−0.026 [−0.050, +0.000]	0.305 (5)	inconclusive
A	RF	S4-Dunford-open [exploratory]		Dunford cat sample		N-target	CC	8	0.1039	0.2274	−0.123 [−0.231, −0.018]	0.305 (5)	inconclusive
A	RF	S4-Dunford-closed [exploratory]		Dunford cat sample		PCC	CC	7	0.0869	0.0942	−0.007 [−0.016, +0.000]	0.344 (5)	inconclusive
A	RF	S4-Dunford-closed [exploratory]		Dunford cat sample		ACC	CC	7	0.0869	0.0965	−0.010 [−0.015, −0.003]	0.156 (5)	inconclusive
A	RF	S4-Dunford-closed [exploratory]		Dunford cat sample		PACC	CC	7	0.0869	0.1023	−0.015 [−0.023, −0.008]	0.156 (5)	inconclusive
A	RF	S4-Dunford-closed [exploratory]		Dunford cat sample		EMQ	CC	7	0.0869	0.1233	−0.036 [−0.079, +0.008]	0.344 (5)	inconclusive
A	RF	S4-Dunford-closed [exploratory]		Dunford cat sample		N-target	CC	7	0.0869	0.2936	−0.207 [−0.371, −0.060]	0.188 (5)	inconclusive
A	RF	S4-Dunford-closed [exploratory]		Dunford cat sample		DM	CC	7	0.0869	0.1101	−0.023 [−0.060, +0.021]	0.688 (3)	inconclusive
A	RF	S4-Dunford-closed [exploratory]		Dunford cat sample		DR-APP	CC	7	0.0869	0.1990	−0.112 [−0.318, +0.067]	0.688 (3)	inconclusive
A	RF	S4-Dunford-closed [exploratory]		Dunford cat sample		DR-natural	CC	7	0.0869	0.2265	−0.140 [−0.325, +0.032]	0.609 (3)	inconclusive
A	RF	R-D [exploratory]		Dunford cat sample		PCC	CC	8	0.0689	0.0756	−0.007 [−0.028, +0.023]	0.891 (5)	inconclusive
A	RF	R-D [exploratory]		Dunford cat sample		ACC	CC	8	0.0689	0.1807	−0.112 [−0.265, −0.001]	0.562 (5)	inconclusive
A	RF	R-D [exploratory]		Dunford cat sample		PACC	CC	8	0.0689	0.1397	−0.071 [−0.186, +0.014]	0.891 (5)	inconclusive
A	RF	R-D [exploratory]		Dunford cat sample		EMQ	CC	8	0.0689	0.1229	−0.054 [−0.108, −0.004]	0.562 (5)	inconclusive
A	RF	R-D [exploratory]		Dunford cat sample		N-target	CC	8	0.0689	0.2274	−0.158 [−0.267, −0.055]	0.156 (5)	inconclusive
A	RF	S1-closed		Smit 30-min (68)		PCC	CC	12	0.0132	0.0235	−0.008 [−0.029, +0.005]	0.947 (4)	inconclusive
A	RF	S1-closed		Smit 30-min (68)		ACC	CC	12	0.0132	0.0189	−0.011 [−0.034, +0.002]	0.701 (4)	inconclusive
A	RF	S1-closed		Smit 30-min (68)		PACC	CC	12	0.0132	0.0564	−0.041 [−0.101, −0.003]	0.092 (4)	inconclusive
A	RF	S1-closed		Smit 30-min (68)		EMQ	CC	12	0.0132	0.1047	−0.080 [−0.207, −0.008]	0.012 (4)	worse
A	RF	S1-L2 (collar)		Smit 30-min (68)		PCC	CC	12	0.0837	0.0870	−0.001 [−0.010, +0.009]	0.807 (4)	equivalent
A	RF	S1-L2 (collar)		Smit 30-min (68)		ACC	CC	12	0.0837	0.1010	−0.020 [−0.029, −0.010]	0.014 (4)	worse
A	RF	S1-L2 (collar)		Smit 30-min (68)		PACC	CC	12	0.0837	0.1044	−0.024 [−0.034, −0.013]	0.014 (4)	worse
A	RF	S1-L2 (collar)		Smit 30-min (68)		EMQ	CC	12	0.0837	0.1041	−0.020 [−0.032, −0.008]	0.015 (4)	worse
A	RF	S1-L2 (collar)		Smit 30-min (68)		N-target	CC	12	0.0837	0.1222	−0.036 [−0.058, −0.015]	0.017 (1)	worse
A	RF	S1-L2 (collar)		Smit 30-min (68)		DM	CC	12	0.0837	0.0966	−0.013 [−0.026, −0.002]	0.192 (3)	inconclusive
A	RF	S1-L2 (collar)		Smit 30-min (68)		DR-APP	CC	12	0.0837	0.1208	−0.029 [−0.063, +0.012]	0.339 (3)	inconclusive
A	RF	S1-L2 (collar)		Smit 30-min (68)		DR-natural	CC	12	0.0837	0.0899	−0.002 [−0.025, +0.028]	0.912 (3)	inconclusive
A	RF	S1-L2 (harness)		Smit 30-min (68)		PCC	CC	12	0.0262	0.0391	−0.014 [−0.018, −0.008]	0.004 (4)	worse
A	RF	S1-L2 (harness)		Smit 30-min (68)		ACC	CC	12	0.0262	0.0430	−0.015 [−0.025, −0.008]	0.004 (4)	worse
A	RF	S1-L2 (harness)		Smit 30-min (68)		PACC	CC	12	0.0262	0.0495	−0.022 [−0.031, −0.014]	0.002 (4)	worse
A	RF	S1-L2 (harness)		Smit 30-min (68)		EMQ	CC	12	0.0262	0.0485	−0.022 [−0.038, −0.008]	0.009 (4)	worse
A	RF	S2-L2 (mapped)		Smit 30-min (68)		PCC	CC	12	0.1573	0.1572	−0.003 [−0.016, +0.009]	0.653 (4)	inconclusive
A	RF	S2-L2 (mapped)		Smit 30-min (68)		ACC	CC	12	0.1573	0.1706	−0.013 [−0.015, −0.009]	0.002 (4)	worse
A	RF	S2-L2 (mapped)		Smit 30-min (68)		PACC	CC	12	0.1573	0.1804	−0.025 [−0.037, −0.014]	0.003 (4)	worse
A	RF	S2-L2 (mapped)		Smit 30-min (68)		EMQ	CC	12	0.1573	0.2141	−0.058 [−0.086, −0.035]	0.003 (4)	worse
A	RF	S1 natural 2-min		Smit 2-min		ACC	CC	12	0.0398	0.0392	−0.001 [−0.004, +0.003]	0.768 (3)	equivalent
A	RF	S1 natural 2-min		Smit 2-min		PACC	CC	12	0.0398	0.0574	−0.018 [−0.043, −0.001]	0.133 (3)	inconclusive

continued

Grp	Clf	Scenario	Labels, features	Samples	Level, m	Arm	Reference	n	AE ref.	AE arm	d [95% CI]	p_{Holm} (family)	Decision
A	RF	S1 natural 2-min		Smit 2-min		EMQ	CC	12	0.0398	0.0891	−0.050 [−0.113, −0.011]	0.009 (3)	worse
A	RF	S1 natural 2-min		Smit 2-min		DM	CC	12	0.0398	0.0678	−0.028 [−0.073, −0.000]	0.126 (3)	inconclusive
A	RF	S1 natural 2-min		Smit 2-min		DR-APP	CC	12	0.0398	0.2685	−0.232 [−0.265, −0.207]	0.001 (3)	worse
A	RF	S1 natural 2-min		Smit 2-min		DR-natural	CC	12	0.0398	0.0586	−0.021 [−0.031, −0.012]	0.004 (3)	worse
A	RF	S1 APP 120-s (bridge)		APP 120-s		ACC	CC	12	0.1522	0.0916	+0.061 [+0.040, +0.078]	0.003 (3)	better
A	RF	S1 APP 120-s (bridge)		APP 120-s		PACC	CC	12	0.1522	0.0921	+0.060 [+0.041, +0.077]	0.003 (3)	better
A	RF	S1 APP 120-s (bridge)		APP 120-s		EMQ	CC	12	0.1522	0.1019	+0.050 [+0.024, +0.075]	0.010 (3)	better
A	RF	S1 APP 120-s (bridge)		APP 120-s		DM	CC	12	0.1522	0.1044	+0.048 [+0.029, +0.066]	0.006 (3)	better
A	RF	S1 APP 120-s (bridge)		APP 120-s		DR-APP	CC	12	0.1522	0.1181	+0.034 [+0.011, +0.054]	0.019 (3)	better
A	RF	S1 APP 120-s (bridge)		APP 120-s		DR-natural	CC	12	0.1522	0.2589	−0.107 [−0.128, −0.084]	0.001 (3)	worse
A	RF	K-L1 within-dog		APP 120-s		PCC	CC	40	0.0133	0.0204	−0.007 [−0.012, −0.003]	0.002 (5)*	inconclusive
A	RF	K-L1 within-dog		APP 120-s		ACC	CC	40	0.0133	0.0166	−0.003 [−0.005, −0.002]	0.001 (5)*	equivalent
A	RF	K-L1 within-dog		APP 120-s		PACC	CC	40	0.0133	0.0231	−0.010 [−0.013, −0.007]	2.5×10^{-5} (5)*	inconclusive
A	RF	K-L1 within-dog		APP 120-s		EMQ	CC	40	0.0133	0.0159	−0.003 [−0.007, +0.001]	0.232 (5)*	equivalent
A	RF	K-L1 within-dog		APP 120-s		N-target	CC	40	0.0133	0.2506	−0.237 [−0.262, −0.214]	2.5×10^{-5} (5)*	worse
A	RF	K-L1 within-dog		dog recordings		PCC	CC	44	0.0114	0.0190	−0.008 [−0.015, −0.002]	0.022 (5)*	inconclusive
A	RF	K-L1 within-dog		dog recordings		ACC	CC	44	0.0114	0.0146	−0.003 [−0.005, −0.002]	0.002 (5)*	equivalent
A	RF	K-L1 within-dog		dog recordings		PACC	CC	44	0.0114	0.0224	−0.011 [−0.017, −0.006]	2.0×10^{-4} (5)*	worse
A	RF	K-L1 within-dog		dog recordings		EMQ	CC	44	0.0114	0.0172	−0.006 [−0.010, −0.002]	0.014 (5)*	equivalent
A	RF	K-L1 within-dog		dog recordings		N-target	CC	44	0.0114	0.2112	−0.200 [−0.231, −0.171]	2.5×10^{-5} (5)*	worse
A	RF	K-L1 within-dog		dog recordings		DM	CC	44	0.0107	0.0152	−0.005 [−0.009, −0.000]	0.039 (3)*	equivalent
A	RF	K-L1 within-dog		dog recordings		DR-APP	CC	44	0.0107	0.0767	−0.066 [−0.084, −0.050]	1.5×10^{-5} (3)*	worse
A	RF	K-L1 within-dog		dog recordings		DR-natural	CC	44	0.0107	0.1320	−0.122 [−0.154, −0.092]	1.5×10^{-5} (3)*	worse
A	RF	K-L1 within-dog		dog 2-min		PCC	CC	35	0.0118	0.0265	−0.015 [−0.024, −0.008]	2.5×10^{-5} (5)*	worse
A	RF	K-L1 within-dog		dog 2-min		ACC	CC	35	0.0118	0.0126	−0.001 [−0.002, +0.001]	0.296 (5)*	equivalent
A	RF	K-L1 within-dog		dog 2-min		PACC	CC	35	0.0118	0.0217	−0.010 [−0.018, −0.004]	4.8×10^{-4} (5)*	inconclusive
A	RF	K-L1 within-dog		dog 2-min		EMQ	CC	35	0.0118	0.0024	+0.009 [+0.002, +0.019]	0.044 (5)*	inconclusive
A	RF	K-L1 within-dog		dog 2-min		N-target	CC	35	0.0118	0.4189	−0.407 [−0.450, −0.366]	2.5×10^{-5} (5)*	worse
A	RF	K-L1 within-dog		dog 2-min		DM	CC	35	0.0078	0.0033	+0.009 [+0.002, +0.018]	0.048 (3)*	inconclusive
A	RF	K-L1 within-dog		dog 2-min		DR-APP	CC	35	0.0078	0.1198	−0.113 [−0.137, −0.093]	1.5×10^{-5} (3)*	worse
A	RF	K-L1 within-dog		dog 2-min		DR-natural	CC	35	0.0078	0.0684	−0.072 [−0.100, −0.047]	1.5×10^{-5} (3)*	worse
A	RF	K-L2 within-dog		APP 120-s		PCC	CC	38	0.0677	0.0693	−0.002 [−0.006, +0.003]	0.548 (5)*	equivalent
A	RF	K-L2 within-dog		APP 120-s		ACC	CC	38	0.0677	0.1040	−0.036 [−0.050, −0.023]	4.0×10^{-5} (5)*	worse
A	RF	K-L2 within-dog		APP 120-s		PACC	CC	38	0.0677	0.1032	−0.036 [−0.048, −0.023]	4.0×10^{-5} (5)*	worse
A	RF	K-L2 within-dog		APP 120-s		EMQ	CC	38	0.0677	0.0903	−0.023 [−0.032, −0.013]	8.0×10^{-5} (5)*	worse
A	RF	K-L2 within-dog		APP 120-s		N-target	CC	38	0.0677	0.1470	−0.079 [−0.094, −0.064]	2.5×10^{-5} (5)*	worse
A	RF	K-L2 within-dog		dog recordings		PCC	CC	44	0.0533	0.0556	−0.002 [−0.008, +0.003]	0.449 (5)*	equivalent
A	RF	K-L2 within-dog		dog recordings		ACC	CC	44	0.0533	0.0845	−0.031 [−0.043, −0.019]	3.0×10^{-5} (5)*	worse
A	RF	K-L2 within-dog		dog recordings		PACC	CC	44	0.0533	0.0840	−0.031 [−0.041, −0.021]	2.5×10^{-5} (5)*	worse
A	RF	K-L2 within-dog		dog recordings		EMQ	CC	44	0.0533	0.0746	−0.021 [−0.030, −0.012]	4.0×10^{-5} (5)*	worse

continued

Grp	Clf	Scenario	Labels, features	Samples	Level, m	Arm	Reference	n	AE ref.	AE arm	d [95% CI]	p_{Holm} (family)	Decision
A	RF	K-L2 within-dog		dog recordings		N-target	CC	44	0.0533	0.1166	−0.063 [−0.079, −0.049]	2.5×10^{-5} (5)*	worse
A	RF	K-L2 within-dog		dog recordings		DM	CC	44	0.0524	0.0797	−0.029 [−0.041, −0.018]	1.5×10^{-5} (3)*	worse
A	RF	K-L2 within-dog		dog recordings		DR-APP	CC	44	0.0524	0.0941	−0.045 [−0.061, −0.031]	1.5×10^{-5} (3)*	worse
A	RF	K-L2 within-dog		dog recordings		DR-natural	CC	44	0.0524	0.1112	−0.058 [−0.074, −0.042]	1.5×10^{-5} (3)*	worse
A	RF	K-L2 within-dog		dog 2-min		PCC	CC	35	0.0880	0.0987	−0.011 [−0.020, −0.001]	0.168 (5)*	inconclusive
A	RF	K-L2 within-dog		dog 2-min		ACC	CC	35	0.0880	0.0969	−0.009 [−0.023, +0.005]	0.426 (5)*	inconclusive
A	RF	K-L2 within-dog		dog 2-min		PACC	CC	35	0.0880	0.0984	−0.010 [−0.023, +0.002]	0.369 (5)*	inconclusive
A	RF	K-L2 within-dog		dog 2-min		EMQ	CC	35	0.0880	0.0885	−0.001 [−0.008, +0.007]	0.897 (5)*	equivalent
A	RF	K-L2 within-dog		dog 2-min		N-target	CC	35	0.0880	0.2834	−0.195 [−0.222, −0.163]	2.5×10^{-5} (5)*	worse
A	RF	K-L2 within-dog		dog 2-min		DM	CC	35	0.0850	0.0772	+0.007 [−0.008, +0.022]	0.412 (3)*	inconclusive
A	RF	K-L2 within-dog		dog 2-min		DR-APP	CC	35	0.0850	0.1927	−0.107 [−0.137, −0.077]	1.5×10^{-5} (3)*	worse
A	RF	K-L2 within-dog		dog 2-min		DR-natural	CC	35	0.0850	0.1358	−0.054 [−0.078, −0.029]	4.7×10^{-4} (3)*	worse
B	RF	within-dog	K-L1, D17	APP 120-s		PCC	CC	40	0.0126	0.0182	−0.006 [−0.009, −0.002]	0.002 (5)*	equivalent
B	RF	within-dog	K-L1, D17	APP 120-s		ACC	CC	40	0.0126	0.0146	−0.002 [−0.004, −0.000]	0.057 (5)*	equivalent
B	RF	within-dog	K-L1, D17	APP 120-s		PACC	CC	40	0.0126	0.0189	−0.006 [−0.009, −0.004]	8.0×10^{-5} (5)*	equivalent
B	RF	within-dog	K-L1, D17	APP 120-s		EMQ	CC	40	0.0126	0.0144	−0.002 [−0.005, +0.001]	0.260 (5)*	equivalent
B	RF	within-dog	K-L1, D17	APP 120-s		N-target	CC	40	0.0126	0.2498	−0.237 [−0.261, −0.214]	2.5×10^{-5} (5)*	worse
B	RF	within-dog	K-L1, D17	dog 2-min		PCC	CC	35	0.0090	0.0212	−0.012 [−0.018, −0.007]	2.5×10^{-5} (5)*	worse
B	RF	within-dog	K-L1, D17	dog 2-min		ACC	CC	35	0.0090	0.0099	−0.001 [−0.002, +0.000]	0.253 (5)*	equivalent
B	RF	within-dog	K-L1, D17	dog 2-min		PACC	CC	35	0.0090	0.0173	−0.008 [−0.014, −0.004]	4.5×10^{-5} (5)*	inconclusive
B	RF	within-dog	K-L1, D17	dog 2-min		EMQ	CC	35	0.0090	0.0044	+0.005 [+0.001, +0.009]	0.071 (5)*	equivalent
B	RF	within-dog	K-L1, D17	dog 2-min		N-target	CC	35	0.0090	0.4189	−0.410 [−0.451, −0.370]	2.5×10^{-5} (5)*	worse
B	RF	within-dog	K-L1, D17	dog recordings		PCC	CC	44	0.0118	0.0146	−0.003 [−0.006, +0.001]	0.114 (5)*	equivalent
B	RF	within-dog	K-L1, D17	dog recordings		ACC	CC	44	0.0118	0.0135	−0.002 [−0.003, −0.000]	0.104 (5)*	equivalent
B	RF	within-dog	K-L1, D17	dog recordings		PACC	CC	44	0.0118	0.0176	−0.006 [−0.009, −0.003]	2.0×10^{-4} (5)*	equivalent
B	RF	within-dog	K-L1, D17	dog recordings		EMQ	CC	44	0.0118	0.0154	−0.004 [−0.007, −0.000]	0.104 (5)*	equivalent
B	RF	within-dog	K-L1, D17	dog recordings		N-target	CC	44	0.0118	0.2112	−0.199 [−0.231, −0.171]	2.5×10^{-5} (5)*	worse
B	RF	within-dog	K-L2, D17	APP 120-s		PCC	CC	38	0.0676	0.0669	+0.001 [−0.006, +0.007]	0.843 (5)*	equivalent
B	RF	within-dog	K-L2, D17	APP 120-s		ACC	CC	38	0.0676	0.0922	−0.025 [−0.034, −0.015]	2.5×10^{-5} (5)*	worse
B	RF	within-dog	K-L2, D17	APP 120-s		PACC	CC	38	0.0676	0.1002	−0.033 [−0.047, −0.021]	2.5×10^{-5} (5)*	worse
B	RF	within-dog	K-L2, D17	APP 120-s		EMQ	CC	38	0.0676	0.0862	−0.019 [−0.026, −0.012]	2.5×10^{-5} (5)*	worse
B	RF	within-dog	K-L2, D17	APP 120-s		N-target	CC	38	0.0676	0.1462	−0.079 [−0.093, −0.066]	2.5×10^{-5} (5)*	worse
B	RF	within-dog	K-L2, D17	dog 2-min		PCC	CC	35	0.0778	0.0956	−0.018 [−0.025, −0.011]	1.0×10^{-4} (5)*	worse
B	RF	within-dog	K-L2, D17	dog 2-min		ACC	CC	35	0.0778	0.0720	+0.006 [−0.004, +0.018]	0.657 (5)*	inconclusive
B	RF	within-dog	K-L2, D17	dog 2-min		PACC	CC	35	0.0778	0.0992	−0.021 [−0.043, −0.003]	0.128 (5)*	inconclusive
B	RF	within-dog	K-L2, D17	dog 2-min		EMQ	CC	35	0.0778	0.0764	+0.001 [−0.011, +0.014]	0.822 (5)*	inconclusive
B	RF	within-dog	K-L2, D17	dog 2-min		N-target	CC	35	0.0778	0.2834	−0.206 [−0.229, −0.180]	2.5×10^{-5} (5)*	worse
B	RF	within-dog	K-L2, D17	dog recordings		PCC	CC	44	0.0500	0.0493	+0.001 [−0.005, +0.007]	0.814 (5)*	equivalent

continued

Grp	Clf	Scenario	Labels, features	Samples	Level, m	Arm	Reference	n	AE ref.	AE arm	d [95% CI]	p_{Holm} (family)	Decision
B	RF	within-dog	K-L2, D17	dog recordings		ACC	CC	44	0.0500	0.0755	−0.026 [−0.037, −0.014]	7.0×10^{-5} (5)*	worse
B	RF	within-dog	K-L2, D17	dog recordings		PACC	CC	44	0.0500	0.0848	−0.035 [−0.047, −0.024]	2.5×10^{-5} (5)*	worse
B	RF	within-dog	K-L2, D17	dog recordings		EMQ	CC	44	0.0500	0.0693	−0.019 [−0.027, −0.012]	2.5×10^{-5} (5)*	worse
B	RF	within-dog	K-L2, D17	dog recordings		N-target	CC	44	0.0500	0.1166	−0.067 [−0.083, −0.050]	2.5×10^{-5} (5)*	worse
B	RF	S2-dog	K-L1, D17	APP 120-s		PCC	CC	40	0.0169	0.0190	−0.002 [−0.004, −0.000]	0.218 (5)*	equivalent
B	RF	S2-dog	K-L1, D17	APP 120-s		ACC	CC	40	0.0169	0.0171	−0.000 [−0.001, +0.000]	0.732 (5)*	equivalent
B	RF	S2-dog	K-L1, D17	APP 120-s		PACC	CC	40	0.0169	0.0181	−0.001 [−0.003, +0.001]	0.706 (5)*	equivalent
B	RF	S2-dog	K-L1, D17	APP 120-s		EMQ	CC	40	0.0169	0.0179	−0.001 [−0.004, +0.003]	0.732 (5)*	equivalent
B	RF	S2-dog	K-L1, D17	APP 120-s		N-target	CC	40	0.0169	0.2514	−0.234 [−0.260, −0.211]	2.5×10^{-5} (5)*	worse
B	RF	S2-dog	K-L1, D17	dog 2-min		PCC	CC	35	0.0103	0.0195	−0.009 [−0.013, −0.006]	2.5×10^{-5} (5)*	inconclusive
B	RF	S2-dog	K-L1, D17	dog 2-min		ACC	CC	35	0.0103	0.0103	−0.000 [−0.000, +0.000]	0.811 (5)*	equivalent
B	RF	S2-dog	K-L1, D17	dog 2-min		PACC	CC	35	0.0103	0.0161	−0.006 [−0.009, −0.003]	3.0×10^{-5} (5)*	equivalent
B	RF	S2-dog	K-L1, D17	dog 2-min		EMQ	CC	35	0.0103	0.0039	+0.006 [+0.002, +0.012]	0.025 (5)*	inconclusive
B	RF	S2-dog	K-L1, D17	dog 2-min		N-target	CC	35	0.0103	0.4189	−0.409 [−0.451, −0.370]	2.5×10^{-5} (5)*	worse
B	RF	S2-dog	K-L1, D17	dog recordings		PCC	CC	44	0.0157	0.0155	+0.000 [−0.002, +0.003]	1.000 (5)*	equivalent
B	RF	S2-dog	K-L1, D17	dog recordings		ACC	CC	44	0.0157	0.0159	−0.000 [−0.001, +0.000]	1.000 (5)*	equivalent
B	RF	S2-dog	K-L1, D17	dog recordings		PACC	CC	44	0.0157	0.0160	−0.000 [−0.003, +0.002]	1.000 (5)*	equivalent
B	RF	S2-dog	K-L1, D17	dog recordings		EMQ	CC	44	0.0157	0.0195	−0.004 [−0.007, −0.000]	0.176 (5)*	equivalent
B	RF	S2-dog	K-L1, D17	dog recordings		N-target	CC	44	0.0157	0.2112	−0.195 [−0.227, −0.166]	2.5×10^{-5} (5)*	worse
C	HGB	S1		Smit 30-min		PCC	CC	12	0.0242	0.0354	−0.011 [−0.016, −0.005]	0.010 (4)	worse
C	HGB	S1		Smit 30-min		ACC	CC	12	0.0242	0.0302	−0.006 [−0.013, −0.000]	0.078 (4)	inconclusive
C	HGB	S1		Smit 30-min		PACC	CC	12	0.0242	0.0406	−0.016 [−0.029, −0.005]	0.037 (4)	worse
C	HGB	S1		Smit 30-min		EMQ	CC	12	0.0242	0.0521	−0.025 [−0.057, −0.003]	0.051 (4)	inconclusive
C	HGB	S2		Smit 30-min		PCC	CC	12	0.0365	0.0535	−0.015 [−0.023, −0.008]	0.008 (4)	worse
C	HGB	S2		Smit 30-min		ACC	CC	12	0.0365	0.0452	−0.009 [−0.017, −0.000]	0.062 (4)	inconclusive
C	HGB	S2		Smit 30-min		PACC	CC	12	0.0365	0.0719	−0.033 [−0.052, −0.014]	0.013 (4)	worse
C	HGB	S2		Smit 30-min		EMQ	CC	12	0.0365	0.1316	−0.088 [−0.160, −0.030]	0.013 (4)	worse
C	HGB	S3 [exploratory]		cat sample		PCC	CC	8	0.1468	0.1698	−0.023 [−0.054, +0.003]	0.562 (4)	inconclusive
C	HGB	S3 [exploratory]		cat sample		ACC	CC	8	0.1468	0.1116	+0.035 [+0.002, +0.082]	0.469 (4)	inconclusive
C	HGB	S3 [exploratory]		cat sample		PACC	CC	8	0.1468	0.1284	+0.018 [−0.019, +0.070]	1.000 (4)	inconclusive
C	HGB	S3 [exploratory]		cat sample		EMQ	CC	8	0.1468	0.1748	−0.028 [−0.118, +0.070]	1.000 (4)	inconclusive
C	HGB	S4-Smit		30-min, open		PCC	CC	12	0.0989	0.0990	−0.000 [−0.000, +0.000]	0.359 (4)	equivalent
C	HGB	S4-Smit		30-min, open		ACC	CC	12	0.0989	0.1002	−0.001 [−0.002, +0.000]	0.359 (4)	equivalent
C	HGB	S4-Smit		30-min, open		PACC	CC	12	0.0989	0.1001	−0.001 [−0.002, +0.000]	0.359 (4)	equivalent
C	HGB	S4-Smit		30-min, open		EMQ	CC	12	0.0989	0.1010	−0.002 [−0.003, −0.000]	0.139 (4)	equivalent
C	HGB	S4-Smit		30-min, closed		PCC	CC	12	0.0091	0.0095	−0.000 [−0.001, −0.000]	0.109 (4)	equivalent
C	HGB	S4-Smit		30-min, closed		ACC	CC	12	0.0091	0.0101	−0.001 [−0.004, +0.002]	1.000 (4)	equivalent
C	HGB	S4-Smit		30-min, closed		PACC	CC	12	0.0091	0.0103	−0.001 [−0.004, +0.002]	1.000 (4)	equivalent
C	HGB	S4-Smit		30-min, closed		EMQ	CC	12	0.0091	0.0114	−0.002 [−0.005, +0.002]	1.000 (4)	equivalent
C	HGB	S4-Dunford [exploratory]		open		PCC	CC	8	0.0982	0.0983	−0.000 [−0.002, +0.002]	0.820 (4)	equivalent
C	HGB	S4-Dunford [exploratory]		open		ACC	CC	8	0.0982	0.1049	−0.007 [−0.010, −0.003]	0.062 (4)	inconclusive
C	HGB	S4-Dunford [exploratory]		open		PACC	CC	8	0.0982	0.1048	−0.007 [−0.011, −0.001]	0.141 (4)	inconclusive
C	HGB	S4-Dunford [exploratory]		open		EMQ	CC	8	0.0982	0.1093	−0.011 [−0.022, +0.003]	0.328 (4)	inconclusive
C	HGB	S4-Dunford [exploratory]		closed		PCC	CC	7	0.0798	0.0799	−0.000 [−0.003, +0.003]	0.953 (4)	equivalent

continued

Grp	Clf	Scenario	Labels, features	Samples	Level, m	Arm	Reference	n	AE ref.	AE arm	d [95% CI]	p_{Holm} (family)	Decision
C	HGB	S4-Dunford [exploratory]		closed		ACC	CC	7	0.0798	0.0888	−0.009 [−0.015, −0.003]	0.125 (4)	inconclusive
C	HGB	S4-Dunford [exploratory]		closed		PACC	CC	7	0.0798	0.0884	−0.009 [−0.016, −0.000]	0.281 (4)	inconclusive
C	HGB	S4-Dunford [exploratory]		closed		EMQ	CC	7	0.0798	0.0949	−0.015 [−0.035, +0.008]	0.531 (4)	inconclusive
C	HGB	S1 bridge (APP vs natural)		APP 120-s		PCC	CC	12	0.1507	0.1710	−0.020 [−0.028, −0.013]	0.002 (4)	worse
C	HGB	S1 bridge (APP vs natural)		APP 120-s		ACC	CC	12	0.1507	0.0914	+0.059 [+0.035, +0.079]	0.003 (4)	better
C	HGB	S1 bridge (APP vs natural)		APP 120-s		PACC	CC	12	0.1507	0.0869	+0.064 [+0.041, +0.082]	0.003 (4)	better
C	HGB	S1 bridge (APP vs natural)		APP 120-s		EMQ	CC	12	0.1507	0.0877	+0.063 [+0.047, +0.078]	0.002 (4)	better
C	HGB	S1 bridge (APP vs natural)		natural 2-min		PCC	CC	12	0.0374	0.0575	−0.020 [−0.025, −0.015]	0.002 (4)	worse
C	HGB	S1 bridge (APP vs natural)		natural 2-min		ACC	CC	12	0.0374	0.0375	−0.001 [−0.007, +0.003]	0.564 (4)	equivalent
C	HGB	S1 bridge (APP vs natural)		natural 2-min		PACC	CC	12	0.0374	0.0480	−0.012 [−0.025, −0.001]	0.126 (4)	inconclusive
C	HGB	S1 bridge (APP vs natural)		natural 2-min		EMQ	CC	12	0.0374	0.0676	−0.032 [−0.061, −0.011]	0.009 (4)	worse
C	LR	S1		Smit 30-min		PCC	CC	12	0.0557	0.0604	−0.008 [−0.016, −0.000]	0.375 (4)	inconclusive
C	LR	S1		Smit 30-min		ACC	CC	12	0.0557	0.0514	+0.005 [−0.014, +0.018]	1.000 (4)	inconclusive
C	LR	S1		Smit 30-min		PACC	CC	12	0.0557	0.0465	+0.007 [−0.007, +0.019]	1.000 (4)	inconclusive
C	LR	S1		Smit 30-min		EMQ	CC	12	0.0557	0.0628	−0.006 [−0.024, +0.007]	1.000 (4)	inconclusive
C	LR	S2		Smit 30-min		PCC	CC	12	0.0533	0.0860	−0.037 [−0.049, −0.027]	0.002 (4)	worse
C	LR	S2		Smit 30-min		ACC	CC	12	0.0533	0.1407	−0.078 [−0.117, −0.043]	0.002 (4)	worse
C	LR	S2		Smit 30-min		PACC	CC	12	0.0533	0.1346	−0.077 [−0.113, −0.041]	0.002 (4)	worse
C	LR	S2		Smit 30-min		EMQ	CC	12	0.0533	0.1753	−0.117 [−0.166, −0.069]	0.002 (4)	worse
C	LR	S3 [exploratory]		cat sample		PCC	CC	8	0.1976	0.1948	+0.003 [−0.055, +0.059]	1.000 (4)	inconclusive
C	LR	S3 [exploratory]		cat sample		ACC	CC	8	0.1976	0.1529	+0.045 [−0.050, +0.193]	1.000 (4)	inconclusive
C	LR	S3 [exploratory]		cat sample		PACC	CC	8	0.1976	0.1714	+0.026 [−0.082, +0.185]	1.000 (4)	inconclusive
C	LR	S3 [exploratory]		cat sample		EMQ	CC	8	0.1976	0.3010	−0.103 [−0.297, +0.116]	1.000 (4)	inconclusive
C	LR	S4-Smit		30-min, open		PCC	CC	12	0.0992	0.1003	−0.001 [−0.002, −0.001]	0.021 (4)	equivalent
C	LR	S4-Smit		30-min, open		ACC	CC	12	0.0992	0.1002	−0.001 [−0.002, +0.000]	0.297 (4)	equivalent
C	LR	S4-Smit		30-min, open		PACC	CC	12	0.0992	0.1003	−0.001 [−0.002, +0.000]	0.297 (4)	equivalent
C	LR	S4-Smit		30-min, open		EMQ	CC	12	0.0992	0.1014	−0.002 [−0.004, −0.000]	0.218 (4)	equivalent
C	LR	S4-Smit		30-min, closed		PCC	CC	12	0.0115	0.0177	−0.006 [−0.009, −0.004]	0.006 (4)	equivalent
C	LR	S4-Smit		30-min, closed		ACC	CC	12	0.0115	0.0112	+0.001 [−0.003, +0.004]	1.000 (4)	equivalent
C	LR	S4-Smit		30-min, closed		PACC	CC	12	0.0115	0.0123	−0.000 [−0.004, +0.003]	1.000 (4)	equivalent
C	LR	S4-Smit		30-min, closed		EMQ	CC	12	0.0115	0.0118	+0.000 [−0.005, +0.006]	1.000 (4)	equivalent
C	LR	S4-Dunford [exploratory]		open		PCC	CC	8	0.0770	0.0816	−0.005 [−0.025, +0.011]	1.000 (4)	inconclusive
C	LR	S4-Dunford [exploratory]		open		ACC	CC	8	0.0770	0.0800	−0.003 [−0.007, +0.001]	0.938 (4)	equivalent
C	LR	S4-Dunford [exploratory]		open		PACC	CC	8	0.0770	0.0804	−0.003 [−0.016, +0.007]	1.000 (4)	inconclusive
C	LR	S4-Dunford [exploratory]		open		EMQ	CC	8	0.0770	0.0891	−0.012 [−0.033, +0.012]	1.000 (4)	inconclusive
C	LR	S4-Dunford [exploratory]		closed		PCC	CC	7	0.0469	0.0547	−0.008 [−0.043, +0.019]	1.000 (4)	inconclusive
C	LR	S4-Dunford [exploratory]		closed		ACC	CC	7	0.0469	0.0521	−0.005 [−0.012, +0.002]	0.875 (4)	inconclusive
C	LR	S4-Dunford [exploratory]		closed		PACC	CC	7	0.0469	0.0525	−0.006 [−0.028, +0.013]	1.000 (4)	inconclusive
C	LR	S4-Dunford [exploratory]		closed		EMQ	CC	7	0.0469	0.0630	−0.016 [−0.051, +0.022]	1.000 (4)	inconclusive
C	LR	S1 bridge (APP vs natural)		APP 120-s		PCC	CC	12	0.2477	0.2087	+0.039 [+0.035, +0.044]	0.002 (4)	better

continued

Grp	Clf	Scenario	Labels, features	Samples	Level, m	Arm	Reference	n	AE ref.	AE arm	d [95% CI]	p_{Holm} (family)	Decision
C	LR	S1 bridge (APP vs natural)		APP 120-s		ACC	CC	12	0.2477	0.1570	+0.091 [+0.059, +0.118]	0.002 (4)	better
C	LR	S1 bridge (APP vs natural)		APP 120-s		PACC	CC	12	0.2477	0.0951	+0.153 [+0.125, +0.176]	0.002 (4)	better
C	LR	S1 bridge (APP vs natural)		APP 120-s		EMQ	CC	12	0.2477	0.1102	+0.138 [+0.109, +0.162]	0.002 (4)	better
C	LR	S1 bridge (APP vs natural)		natural 2-min		PCC	CC	12	0.0740	0.0969	−0.023 [−0.028, −0.018]	0.004 (4)	worse
C	LR	S1 bridge (APP vs natural)		natural 2-min		ACC	CC	12	0.0740	0.0802	−0.005 [−0.023, +0.009]	1.000 (4)	inconclusive
C	LR	S1 bridge (APP vs natural)		natural 2-min		PACC	CC	12	0.0740	0.0644	+0.008 [−0.009, +0.024]	1.000 (4)	inconclusive
C	LR	S1 bridge (APP vs natural)		natural 2-min		EMQ	CC	12	0.0740	0.0732	−0.001 [−0.021, +0.019]	1.000 (4)	inconclusive
D	RF	S1		2-min		DR-APP-h	CC	12	0.0398	0.0902	−0.050 [−0.101, −0.019]	9.8×10^{-4} (2)	worse
D	RF	S1		2-min		DR-natural-h	CC	12	0.0398	0.0602	−0.020 [−0.049, −0.002]	0.033 (2)	worse
D	RF	S1		30-min		DR-APP-h	CC	12	0.0260	0.0784	−0.044 [−0.096, −0.012]	0.002 (2)	worse
D	RF	S1		30-min		DR-natural-h	CC	12	0.0260	0.0787	−0.043 [−0.071, −0.018]	0.003 (2)	worse
D	RF	S1		APP 120-s		DR-APP-h	CC	12	0.1522	0.1043	+0.048 [+0.031, +0.065]	0.003 (2)	better
D	RF	S1		APP 120-s		DR-natural-h	CC	12	0.1522	0.1662	−0.014 [−0.030, +0.004]	0.147 (2)	inconclusive
D	RF	S1-L2 (collar)		30-min		DR-APP-h	CC	12	0.0837	0.1167	−0.034 [−0.056, −0.009]	0.057 (2)	inconclusive
D	RF	S1-L2 (collar)		30-min		DR-natural-h	CC	12	0.0837	0.0915	−0.009 [−0.022, +0.003]	0.226 (2)	inconclusive
D	RF	S2		30-min		DR-APP-h	CC	12	0.0405	0.1600	−0.107 [−0.201, −0.032]	0.009 (2)	worse
D	RF	S2		30-min		DR-natural-h	CC	12	0.0405	0.1051	−0.054 [−0.117, −0.007]	0.054 (2)	inconclusive
D	RF	S4-Smit-closed		30-min		DR-APP-h	CC	12	0.0089	0.0334	−0.023 [−0.029, −0.016]	0.002 (2)	worse
D	RF	S4-Smit-closed		30-min		DR-natural-h	CC	12	0.0089	0.0267	−0.016 [−0.022, −0.010]	0.002 (2)	worse
D	RF	K-L1		natural 2-min		DR-APP-h	CC	35	0.0078	0.0475	−0.039 [−0.042, −0.034]	1.0×10^{-5} (2)*	worse
D	RF	K-L1		natural 2-min		DR-natural-h	CC	35	0.0078	0.0314	−0.017 [−0.030, −0.006]	0.009 (2)*	worse
D	RF	K-L1		recording		DR-APP-h	CC	44	0.0107	0.0446	−0.032 [−0.043, −0.023]	1.0×10^{-5} (2)*	worse
D	RF	K-L1		recording		DR-natural-h	CC	44	0.0107	0.0848	−0.070 [−0.085, −0.057]	1.0×10^{-5} (2)*	worse
D	RF	K-L2		natural 2-min		DR-APP-h	CC	35	0.0850	0.1217	−0.022 [−0.042, −0.002]	0.099 (2)*	inconclusive
D	RF	K-L2		natural 2-min		DR-natural-h	CC	35	0.0850	0.1012	−0.004 [−0.024, +0.015]	0.681 (2)*	inconclusive
D	RF	K-L2		recording		DR-APP-h	CC	44	0.0524	0.0760	−0.022 [−0.033, −0.009]	0.001 (2)*	worse
D	RF	K-L2		recording		DR-natural-h	CC	44	0.0524	0.0860	−0.032 [−0.044, −0.020]	3.0×10^{-5} (2)*	worse
D	RF	S3 [exploratory]		cat sample		DR-APP-h	CC	8	0.1442	0.1519	−0.008 [−0.067, +0.051]	0.812 (2)	inconclusive
D	RF	S3 [exploratory]		cat sample		DR-natural-h	CC	8	0.1442	0.1171	+0.027 [−0.014, +0.070]	0.641 (2)	inconclusive
D	RF	S4-Dunford-closed [exploratory]		cat sample		DR-APP-h	CC	7	0.0869	0.1385	−0.052 [−0.089, −0.014]	0.125 (2)	inconclusive
D	RF	S4-Dunford-closed [exploratory]		cat sample		DR-natural-h	CC	7	0.0869	0.1379	−0.051 [−0.141, +0.032]	0.391 (2)	inconclusive
E	RF	S3 [exploratory]		-	study, m = 300	Retraining	CC	8	0.1442	0.1468	−0.003 [−0.006, +0.000]	0.562 (2)	equivalent
E	RF	S3 [exploratory]		-	study, m = 1,800	Retraining	CC	8	0.1442	0.1404	+0.004 [−0.002, +0.010]	0.562 (2)	inconclusive
E	RF	S3 [exploratory]		-	study, m = 300	Weighted retraining	CC	8	0.1442	0.1510	−0.007 [−0.015, +0.000]	0.391 (2)	inconclusive
E	RF	S3 [exploratory]		-	study, m = 1,800	Weighted retraining	CC	8	0.1442	0.1464	−0.002 [−0.012, +0.006]	0.672 (2)	inconclusive
E	RF	S3 [exploratory]		-	study, m = 300	Weighted retraining	Retraining	8	0.1468	0.1510	−0.004 [−0.010, +0.001]	0.328 (2)	equivalent
E	RF	S3 [exploratory]		-	study, m = 1,800	Weighted retraining	Retraining	8	0.1404	0.1464	−0.006 [−0.013, +0.001]	0.328 (2)	inconclusive
E	RF	S4-Dunford [exploratory]		open	study, m = 300	Retraining	CC	8	0.1039	0.0916	+0.012 [+0.003, +0.023]	0.078 (2)	inconclusive

continued

Grp	Clf	Scenario	Labels, features	Samples	Level, m	Arm	Reference	n	AE ref.	AE arm	d [95% CI]	p_{Holm} (family)	Decision
E	RF	S4-Dunford [exploratory]		open	study, m = 1,800	Retraining	CC	8	0.1039	0.0831	+0.021 [+0.007, +0.038]	0.078 (2)	inconclusive
E	RF	S4-Dunford [exploratory]		open	study, m = 300	Weighted retraining	CC	8	0.1039	0.0916	+0.012 [+0.003, +0.023]	0.078 (2)	inconclusive
E	RF	S4-Dunford [exploratory]		open	study, m = 1,800	Weighted retraining	CC	8	0.1039	0.0826	+0.021 [+0.007, +0.039]	0.078 (2)	inconclusive
E	RF	S4-Dunford [exploratory]		open	study, m = 300	Weighted retraining	Retraining	8	0.0916	0.0916	+0.000 [−0.001, +0.001]	1.000 (2)	equivalent
E	RF	S4-Dunford [exploratory]		open	study, m = 1,800	Weighted retraining	Retraining	8	0.0831	0.0826	+0.001 [−0.001, +0.002]	1.000 (2)	equivalent
E	RF	S4-Dunford [exploratory]		closed	study, m = 300	Retraining	CC	7	0.0869	0.0729	+0.014 [+0.003, +0.028]	0.125 (2)	inconclusive
E	RF	S4-Dunford [exploratory]		closed	study, m = 1,800	Retraining	CC	7	0.0869	0.0591	+0.028 [+0.005, +0.056]	0.125 (2)	inconclusive
E	RF	S4-Dunford [exploratory]		closed	study, m = 300	Weighted retraining	CC	7	0.0869	0.0761	+0.011 [+0.003, +0.020]	0.156 (2)	inconclusive
E	RF	S4-Dunford [exploratory]		closed	study, m = 1,800	Weighted retraining	CC	7	0.0869	0.0573	+0.030 [+0.006, +0.058]	0.156 (2)	inconclusive
E	RF	S4-Dunford [exploratory]		closed	study, m = 300	Weighted retraining	Retraining	7	0.0729	0.0761	−0.003 [−0.008, +0.000]	0.250 (2)	equivalent
E	RF	S4-Dunford [exploratory]		closed	study, m = 1,800	Weighted retraining	Retraining	7	0.0591	0.0573	+0.002 [+0.000, +0.003]	0.250 (2)	equivalent
E	RF	S4-Smit		open	individual, m = 60	Retraining	CC	12	0.0991	0.1050	−0.005 [−0.014, +0.000]	0.535 (4)	inconclusive
E	RF	S4-Smit		open	individual, m = 300	Retraining	CC	12	0.0990	0.0951	+0.003 [−0.002, +0.009]	0.535 (4)	equivalent
E	RF	S4-Smit		open	individual, m = 900	Retraining	CC	12	0.1015	0.0917	+0.009 [+0.003, +0.017]	0.006 (4)	inconclusive
E	RF	S4-Smit		open	individual, m = 1,800	Retraining	CC	12	0.1021	0.0828	+0.017 [+0.007, +0.027]	0.006 (4)	better
E	RF	S4-Smit		open	individual, m = 300	Weighted retraining	CC	12	0.0990	0.0953	+0.003 [−0.000, +0.007]	0.138 (2)	equivalent
E	RF	S4-Smit		open	individual, m = 1,800	Weighted retraining	CC	12	0.1021	0.0857	+0.014 [+0.006, +0.024]	0.006 (2)	better
E	RF	S4-Smit		open	individual, m = 300	Weighted retraining	Retraining	12	0.0951	0.0953	−0.000 [−0.002, +0.002]	0.934 (2)	equivalent
E	RF	S4-Smit		open	individual, m = 1,800	Weighted retraining	Retraining	12	0.0828	0.0857	−0.003 [−0.004, −0.001]	0.025 (2)	equivalent
E	RF	S4-Smit		open	study, m = 60	Retraining	CC	12	0.0991	0.0990	+0.000 [−0.000, +0.000]	1.000 (4)	equivalent
E	RF	S4-Smit		open	study, m = 300	Retraining	CC	12	0.0990	0.0992	−0.001 [−0.007, +0.004]	1.000 (4)	equivalent
E	RF	S4-Smit		open	study, m = 900	Retraining	CC	12	0.1015	0.0990	+0.002 [+0.001, +0.004]	0.019 (4)	equivalent
E	RF	S4-Smit		open	study, m = 1,800	Retraining	CC	12	0.1021	0.0944	+0.007 [+0.003, +0.013]	0.006 (4)	inconclusive
E	RF	S4-Smit		open	study, m = 300	Weighted retraining	CC	12	0.0990	0.0967	+0.002 [+0.001, +0.004]	0.020 (2)	equivalent
E	RF	S4-Smit		open	study, m = 1,800	Weighted retraining	CC	12	0.1021	0.0959	+0.005 [+0.002, +0.010]	0.007 (2)	equivalent
E	RF	S4-Smit		open	study, m = 300	Weighted retraining	Retraining	12	0.0992	0.0967	+0.003 [−0.001, +0.009]	0.939 (2)	equivalent
E	RF	S4-Smit		open	study, m = 1,800	Weighted retraining	Retraining	12	0.0944	0.0959	−0.001 [−0.003, −0.000]	0.014 (2)	equivalent
E	RF	S1		Smit 30-min	individual, m = 300	Retraining	CC	12	0.0264	0.0244	+0.002 [−0.000, +0.005]	1.000 (2)	equivalent
E	RF	S1		Smit 30-min	individual, m = 1,800	Retraining	CC	12	0.0293	0.0264	+0.002 [−0.000, +0.007]	1.000 (2)	equivalent
E	RF	S1		Smit 30-min	individual, m = 300	Weighted retraining	CC	12	0.0264	0.0244	+0.002 [−0.000, +0.005]	1.000 (2)	equivalent

continued

Grp	Clf	Scenario	Labels, features	Samples	Level, m	Arm	Reference	n	AE ref.	AE arm	d [95% CI]	p_{Holm} (family)	Decision
E	RF	S1		Smit 30-min	individual, m = 1,800	Weighted retraining	CC	12	0.0293	0.0267	+0.002 [−0.001, +0.007]	1.000 (2)	equivalent
E	RF	S1		Smit 30-min	individual, m = 300	Weighted retraining	Retraining	12	0.0244	0.0244	−0.000 [−0.000, +0.000]	0.382 (2)	equivalent
E	RF	S1		Smit 30-min	individual, m = 1,800	Weighted retraining	Retraining	12	0.0264	0.0267	−0.000 [−0.001, −0.000]	0.100 (2)	equivalent
E	RF	S2		Smit 30-min	individual, m = 300	Retraining	CC	12	0.0410	0.0356	+0.005 [+0.001, +0.010]	9.8×10^{-4} (2)	equivalent
E	RF	S2		Smit 30-min	individual, m = 1,800	Retraining	CC	12	0.0436	0.0359	+0.007 [+0.003, +0.012]	9.8×10^{-4} (2)	inconclusive
E	RF	S2		Smit 30-min	individual, m = 300	Weighted retraining	CC	12	0.0410	0.0358	+0.005 [+0.001, +0.009]	0.006 (2)	equivalent
E	RF	S2		Smit 30-min	individual, m = 1,800	Weighted retraining	CC	12	0.0436	0.0360	+0.007 [+0.003, +0.012]	0.002 (2)	inconclusive
E	RF	S2		Smit 30-min	individual, m = 300	Weighted retraining	Retraining	12	0.0356	0.0358	−0.000 [−0.001, +0.000]	1.000 (2)	equivalent
E	RF	S2		Smit 30-min	individual, m = 1,800	Weighted retraining	Retraining	12	0.0359	0.0360	−0.000 [−0.001, +0.000]	1.000 (2)	equivalent
F	RF	S1		Smit 30-min	individual, m = 1,800	T-ACC, stratified (gain over own CC)	T-ACC, random blocks (gain over own CC)	12	0.0627	0.0242	+0.046 [+0.020, +0.083]	9.8×10^{-4} (2)	better
F	RF	S1		Smit 30-min	individual, m = 300	PACC-target, stratified (gain over own CC)	PACC-target, random blocks (gain over own CC)	12	0.0571	0.0390	+0.018 [+0.012, +0.025]	9.8×10^{-4} (2)	better
F	RF	S1		Smit 30-min	individual, m = 1,800	PACC-target, stratified (gain over own CC)	PACC-target, random blocks (gain over own CC)	12	0.0669	0.0390	+0.033 [+0.020, +0.050]	9.8×10^{-4} (2)	better
F	RF	S2		Smit 30-min	individual, m = 1,800	T-ACC, stratified (gain over own CC)	T-ACC, random blocks (gain over own CC)	12	0.0723	0.0438	+0.038 [+0.019, +0.060]	0.002 (2)	better
F	RF	S2		Smit 30-min	individual, m = 300	PACC-target, stratified (gain over own CC)	PACC-target, random blocks (gain over own CC)	12	0.0893	0.0689	+0.021 [+0.015, +0.027]	9.8×10^{-4} (2)	better
F	RF	S2		Smit 30-min	individual, m = 1,800	PACC-target, stratified (gain over own CC)	PACC-target, random blocks (gain over own CC)	12	0.0877	0.0663	+0.027 [+0.014, +0.040]	0.001 (2)	better
F	RF	S3 [exploratory]		Dunford cat sample	study, m = 1,800	T-ACC, stratified (gain over own CC)	T-ACC, random blocks (gain over own CC)	8	0.1414	0.1502	−0.009 [−0.026, +0.005]	0.797 (4)	inconclusive
F	RF	S3 [exploratory]		Dunford cat sample	study, m = 60	PACC-target, stratified (gain over own CC)	PACC-target, random blocks (gain over own CC)	8	0.1150	0.1747	−0.060 [−0.136, +0.023]	0.875 (4)	inconclusive
F	RF	S3 [exploratory]		Dunford cat sample	study, m = 300	PACC-target, stratified (gain over own CC)	PACC-target, random blocks (gain over own CC)	8	0.1119	0.1527	−0.041 [−0.149, +0.034]	1.000 (4)	inconclusive
F	RF	S3 [exploratory]		Dunford cat sample	study, m = 900	PACC-target, stratified (gain over own CC)	PACC-target, random blocks (gain over own CC)	8	0.1226	0.1508	−0.028 [−0.117, +0.035]	1.000 (4)	inconclusive
F	RF	S3 [exploratory]		Dunford cat sample	study, m = 1,800	PACC-target, stratified (gain over own CC)	PACC-target, random blocks (gain over own CC)	8	0.1221	0.1511	−0.029 [−0.087, +0.005]	1.000 (4)	inconclusive
F	RF	S4-Dunford-open [exploratory]		Dunford cat sample	study, m = 1,800	T-ACC, stratified (gain over own CC)	T-ACC, random blocks (gain over own CC)	8	0.0928	0.0972	−0.004 [−0.023, +0.007]	1.000 (4)	inconclusive
F	RF	S4-Dunford-open [exploratory]		Dunford cat sample	study, m = 60	PACC-target, stratified (gain over own CC)	PACC-target, random blocks (gain over own CC)	8	0.0975	0.0967	+0.001 [−0.026, +0.017]	1.000 (4)	inconclusive
F	RF	S4-Dunford-open [exploratory]		Dunford cat sample	study, m = 300	PACC-target, stratified (gain over own CC)	PACC-target, random blocks (gain over own CC)	8	0.0935	0.0947	−0.001 [−0.024, +0.014]	1.000 (4)	inconclusive
F	RF	S4-Dunford-open [exploratory]		Dunford cat sample	study, m = 900	PACC-target, stratified (gain over own CC)	PACC-target, random blocks (gain over own CC)	8	0.0922	0.0936	−0.001 [−0.017, +0.009]	1.000 (4)	inconclusive

continued

Grp	Clf	Scenario	Labels, features	Samples	Level, m	Arm	Reference	n	AE ref.	AE arm	d [95% CI]	p_{Holm} (family)	Decision
F	RF	S4-Dunford-open [exploratory]		Dunford cat sample	study, m = 1,800	PACC-target, stratified (gain over own CC)	PACC-target, random blocks (gain over own CC)	8	0.0919	0.0938	−0.002 [−0.018, +0.009]	1.000 (4)	inconclusive
F	RF	S4-Dunford-closed [exploratory]		Dunford cat sample	study, m = 1,800	T-ACC, stratified (gain over own CC)	T-ACC, random blocks (gain over own CC)	7	0.0815	0.0906	−0.009 [−0.042, +0.011]	1.000 (4)	inconclusive
F	RF	S4-Dunford-closed [exploratory]		Dunford cat sample	study, m = 60	PACC-target, stratified (gain over own CC)	PACC-target, random blocks (gain over own CC)	7	0.0956	0.0843	+0.011 [−0.033, +0.055]	1.000 (4)	inconclusive
F	RF	S4-Dunford-closed [exploratory]		Dunford cat sample	study, m = 300	PACC-target, stratified (gain over own CC)	PACC-target, random blocks (gain over own CC)	7	0.0834	0.0827	+0.001 [−0.025, +0.020]	1.000 (4)	inconclusive
F	RF	S4-Dunford-closed [exploratory]		Dunford cat sample	study, m = 900	PACC-target, stratified (gain over own CC)	PACC-target, random blocks (gain over own CC)	7	0.0818	0.0830	−0.001 [−0.027, +0.018]	1.000 (4)	inconclusive
F	RF	S4-Dunford-closed [exploratory]		Dunford cat sample	study, m = 1,800	PACC-target, stratified (gain over own CC)	PACC-target, random blocks (gain over own CC)	7	0.0790	0.0835	−0.005 [−0.031, +0.014]	1.000 (4)	inconclusive
F	RF	S4-Smit-open		Smit 30-min	individual, m = 1,800	T-ACC, stratified (gain over own CC)	T-ACC, random blocks (gain over own CC)	12	0.1113	0.0760	+0.013 [−0.003, +0.037]	0.842 (4)	inconclusive
F	RF	S4-Smit-open		Smit 30-min	individual, m = 60	PACC-target, stratified (gain over own CC)	PACC-target, random blocks (gain over own CC)	12	0.1000	0.1047	−0.006 [−0.012, −0.001]	0.041 (4)	inconclusive
F	RF	S4-Smit-open		Smit 30-min	individual, m = 300	PACC-target, stratified (gain over own CC)	PACC-target, random blocks (gain over own CC)	12	0.1016	0.1009	−0.003 [−0.007, +0.002]	0.861 (4)	equivalent
F	RF	S4-Smit-open		Smit 30-min	individual, m = 900	PACC-target, stratified (gain over own CC)	PACC-target, random blocks (gain over own CC)	12	0.1088	0.0918	+0.003 [−0.003, +0.008]	0.861 (4)	equivalent
F	RF	S4-Smit-open		Smit 30-min	individual, m = 1,800	PACC-target, stratified (gain over own CC)	PACC-target, random blocks (gain over own CC)	12	0.1119	0.0769	+0.013 [−0.003, +0.037]	0.861 (4)	inconclusive
F	RF	S4-Smit-open		Smit 30-min	individual, m = 300	Target-only, stratified (gain over own CC)	Target-only, random blocks (gain over own CC)	12	0.0739	0.0508	+0.015 [−0.003, +0.031]	0.124 (2)	inconclusive
F	RF	S4-Smit-open		Smit 30-min	individual, m = 1,800	Target-only, stratified (gain over own CC)	Target-only, random blocks (gain over own CC)	12	0.0382	0.0536	−0.044 [−0.058, −0.030]	9.8×10^{-4} (2)	worse
F	RF	S4-Smit-open		Smit 30-min	study, m = 60	PACC-target, stratified (gain over own CC)	PACC-target, random blocks (gain over own CC)	12	0.1025	0.1054	−0.005 [−0.010, +0.000]	0.417 (4)	equivalent
F	RF	S4-Smit-open		Smit 30-min	study, m = 300	PACC-target, stratified (gain over own CC)	PACC-target, random blocks (gain over own CC)	12	0.1036	0.1030	−0.001 [−0.005, +0.005]	1.000 (4)	equivalent
F	RF	S4-Smit-open		Smit 30-min	study, m = 900	PACC-target, stratified (gain over own CC)	PACC-target, random blocks (gain over own CC)	12	0.1060	0.1030	+0.001 [−0.002, +0.004]	1.000 (4)	equivalent
F	RF	S4-Smit-open		Smit 30-min	study, m = 1,800	PACC-target, stratified (gain over own CC)	PACC-target, random blocks (gain over own CC)	12	0.1149	0.1028	+0.008 [+0.001, +0.016]	0.213 (4)	inconclusive
F	RF	S4-Smit-closed		Smit 30-min	individual, m = 1,800	T-ACC, stratified (gain over own CC)	T-ACC, random blocks (gain over own CC)	12	0.0266	0.0249	+0.010 [−0.009, +0.034]	1.000 (4)	inconclusive
F	RF	S4-Smit-closed		Smit 30-min	individual, m = 60	PACC-target, stratified (gain over own CC)	PACC-target, random blocks (gain over own CC)	12	0.0098	0.0277	−0.017 [−0.028, −0.009]	0.002 (4)	worse
F	RF	S4-Smit-closed		Smit 30-min	individual, m = 300	PACC-target, stratified (gain over own CC)	PACC-target, random blocks (gain over own CC)	12	0.0187	0.0240	−0.003 [−0.016, +0.010]	1.000 (4)	inconclusive
F	RF	S4-Smit-closed		Smit 30-min	individual, m = 900	PACC-target, stratified (gain over own CC)	PACC-target, random blocks (gain over own CC)	12	0.0257	0.0257	+0.005 [−0.008, +0.022]	1.000 (4)	inconclusive

continued

Grp	Clf	Scenario	Labels, features	Samples	Level, m	Arm	Reference	n	AE ref.	AE arm	d [95% CI]	p_{Holm} (family)	Decision
F	RF	S4-Smit-closed		Smit 30-min	individual, m = 1,800	PACC-target, stratified (gain over own CC)	PACC-target, random blocks (gain over own CC)	12	0.0275	0.0275	+0.008 [−0.009, +0.031]	1.000 (4)	inconclusive
F	RF	S4-Smit-closed		Smit 30-min	study, m = 60	PACC-target, stratified (gain over own CC)	PACC-target, random blocks (gain over own CC)	12	0.0101	0.0263	−0.017 [−0.024, −0.012]	0.002 (4)	worse
F	RF	S4-Smit-closed		Smit 30-min	study, m = 300	PACC-target, stratified (gain over own CC)	PACC-target, random blocks (gain over own CC)	12	0.0163	0.0233	−0.007 [−0.012, −0.003]	0.038 (4)	inconclusive
F	RF	S4-Smit-closed		Smit 30-min	study, m = 900	PACC-target, stratified (gain over own CC)	PACC-target, random blocks (gain over own CC)	12	0.0356	0.0223	+0.008 [−0.004, +0.022]	0.271 (4)	inconclusive
F	RF	S4-Smit-closed		Smit 30-min	study, m = 1,800	PACC-target, stratified (gain over own CC)	PACC-target, random blocks (gain over own CC)	12	0.0270	0.0215	+0.004 [−0.000, +0.009]	0.227 (4)	equivalent

Table S4. Macro-F1, accuracy and pooled top-label expected calibration error (ECE, 10 bins) of the three classifiers.

ECE is pooled over individuals and computed on the highest predicted class probability.

scenario	variant	classifier	n individuals	macroF1	accuracy	ECE pooled
S1	-	HGB	12	0.699	0.905	0.0032
S1	-	LR	12	0.534	0.863	0.0536
S1	-	RF	12	0.702	0.904	0.0181
S2	-	HGB	12	0.643	0.860	0.0075
S2	-	LR	12	0.557	0.852	0.0645
S2	-	RF	12	0.638	0.856	0.0419
S3 [exploratory]	-	HGB	9	0.596	0.689	0.0998
S3 [exploratory]	-	LR	9	0.482	0.615	0.1203
S3 [exploratory]	-	RF	9	0.588	0.695	0.0732
S4-Dunford [exploratory]	open	HGB	9	0.676	0.825	0.1021
S4-Dunford [exploratory]	open	LR	9	0.670	0.834	0.0421
S4-Dunford [exploratory]	open	RF	9	0.677	0.826	0.0834
S4-Dunford [exploratory]	closed	HGB	9	0.819	0.890	0.0759
S4-Dunford [exploratory]	closed	LR	9	0.815	0.896	0.0268
S4-Dunford [exploratory]	closed	RF	9	0.816	0.889	0.0584
S4-Smit	open	HGB	12	0.436	0.818	0.1652
S4-Smit	open	LR	12	0.435	0.818	0.1497
S4-Smit	open	RF	12	0.435	0.818	0.1517
S4-Smit	closed	HGB	12	0.782	0.963	0.0259
S4-Smit	closed	LR	12	0.780	0.962	0.0184
S4-Smit	closed	RF	12	0.781	0.963	0.0200

Table S5. Distribution matching (DM): solutions on the boundary of the simplex and classes set to zero.

share_boundary_solution: share of samples in which DM set at least one class to zero. For each class: number and share of samples in which DM set it to zero, how many of those samples truly lacked the class, and the median [quartiles] of its true proportion in those samples.

scenario	sample type	n samples	share boundary solution	class	n DM zero	share DM zero	n true also zero	share true also zero	true median	true q1	true q3
K-L1	natural 2-min	130	0.754	inactive	67	0.515	67	1.000	0.000	0.000	0.000
K-L1	natural 2-min	130	0.754	locomotion	31	0.238	31	1.000	0.000	0.000	0.000
K-L2	natural 2-min	130	1.000	Lying on chest	95	0.731	83	0.874	0.000	0.000	0.000
K-L2	natural 2-min	130	1.000	Sitting	107	0.823	102	0.953	0.000	0.000	0.000
K-L2	natural 2-min	130	1.000	Standing	116	0.892	102	0.879	0.000	0.000	0.000
K-L2	natural 2-min	130	1.000	Trotting	80	0.615	80	1.000	0.000	0.000	0.000
K-L2	natural 2-min	130	1.000	Walking	77	0.592	73	0.948	0.000	0.000	0.000
K-L2	recording	61	0.984	Lying on chest	21	0.344	0	0.000	0.067	0.014	0.187
K-L2	recording	61	0.984	Sitting	38	0.623	12	0.316	0.021	0.000	0.083
K-L2	recording	61	0.984	Standing	42	0.689	11	0.262	0.021	0.001	0.065
S1	2-min	1,083	0.887	inactive	108	0.100	28	0.259	0.076	0.000	0.707
S1	2-min	1,083	0.887	locomotion	811	0.749	748	0.922	0.000	0.000	0.000
S1	2-min	1,083	0.887	maintenance	723	0.668	675	0.934	0.000	0.000	0.000
S1	30-min	68	0.676	inactive	2	0.029	0	0.000	0.567	0.477	0.656
S1	30-min	68	0.676	locomotion	38	0.559	17	0.447	0.002	0.000	0.013
S1	30-min	68	0.676	maintenance	23	0.338	8	0.348	0.007	0.000	0.022
S1	APP 120-s	600	0.268	inactive	40	0.067	2	0.050	0.042	0.017	0.127
S1	APP 120-s	600	0.268	locomotion	66	0.110	2	0.030	0.050	0.019	0.075
S1	APP 120-s	600	0.268	maintenance	59	0.098	5	0.085	0.100	0.033	0.204
S1-L2 (collar)	30-min	68	1.000	Eating	44	0.647	38	0.864	0.000	0.000	0.000
S1-L2 (collar)	30-min	68	1.000	Grooming	22	0.324	10	0.455	0.001	0.000	0.020
S1-L2 (collar)	30-min	68	1.000	Lying	15	0.221	4	0.267	0.280	0.003	0.461
S1-L2 (collar)	30-min	68	1.000	Sitting	38	0.559	3	0.079	0.116	0.032	0.273
S1-L2 (collar)	30-min	68	1.000	Standing	54	0.794	13	0.241	0.016	0.001	0.072
S1-L2 (collar)	30-min	68	1.000	Trotting	51	0.750	41	0.804	0.000	0.000	0.000
S1-L2 (collar)	30-min	68	1.000	Walking	35	0.515	17	0.486	0.001	0.000	0.011
S2	30-min	68	0.485	inactive	5	0.074	0	0.000	0.605	0.404	0.764
S2	30-min	68	0.485	locomotion	22	0.324	15	0.682	0.000	0.000	0.001
S2	30-min	68	0.485	maintenance	21	0.309	9	0.429	0.002	0.000	0.033
S3 [exploratory]	cat sample	8	0.625	inactive	1	0.125	0	0.000	0.052	0.052	0.052
S3 [exploratory]	cat sample	8	0.625	locomotion	3	0.375	1	0.333	0.004	0.002	0.055
S3 [exploratory]	cat sample	8	0.625	maintenance	4	0.500	3	0.750	0.000	0.000	0.126
S4-Dunford-closed [exploratory]	cat sample	7	0.286	locomotion	2	0.286	1	0.500	0.002	0.001	0.003
S4-Smit-closed	30-min	68	0.500	locomotion	34	0.500	18	0.529	0.000	0.000	0.006

Table S6. Direct compositional regression (DR): diagnostics.

(a) Training samples of DR-natural: share of samples in which a class was absent, and its mean and median true proportion.

training sample type	label space	class	n samples	share true zero	true mean	true median
Smit 30-min (coverage $\geq$ 50%)	L1	inactive	68	0.000	0.821	0.887
Smit 30-min (coverage $\geq$ 50%)	L1	locomotion	68	0.265	0.032	0.014
Smit 30-min (coverage $\geq$ 50%)	L1	maintenance	68	0.147	0.147	0.059
Smit 2-min (coverage $\geq$ 50%)	L1	inactive	1,083	0.029	0.802	1.000
Smit 2-min (coverage $\geq$ 50%)	L1	locomotion	1,083	0.753	0.035	0.000
Smit 2-min (coverage $\geq$ 50%)	L1	maintenance	1,083	0.685	0.164	0.000
Smit 30-min (coverage $\geq$ 50%)	L2	Lying	68	0.191	0.472	0.493
Smit 30-min (coverage $\geq$ 50%)	L2	Sitting	68	0.074	0.284	0.167
Smit 30-min (coverage $\geq$ 50%)	L2	Standing	68	0.191	0.065	0.026
Smit 30-min (coverage $\geq$ 50%)	L2	Walking	68	0.265	0.029	0.014
Smit 30-min (coverage $\geq$ 50%)	L2	Trotting	68	0.750	0.003	0.000
Smit 30-min (coverage $\geq$ 50%)	L2	Grooming	68	0.162	0.103	0.052
Smit 30-min (coverage $\geq$ 50%)	L2	Eating	68	0.750	0.043	0.000
Smit 2-min (coverage $\geq$ 50%)	L2	Lying	1,083	0.492	0.441	0.029
Smit 2-min (coverage $\geq$ 50%)	L2	Sitting	1,083	0.485	0.286	0.017
Smit 2-min (coverage $\geq$ 50%)	L2	Standing	1,083	0.734	0.075	0.000
Smit 2-min (coverage $\geq$ 50%)	L2	Walking	1,083	0.758	0.032	0.000
Smit 2-min (coverage $\geq$ 50%)	L2	Trotting	1,083	0.957	0.003	0.000
Smit 2-min (coverage $\geq$ 50%)	L2	Grooming	1,083	0.763	0.104	0.000
Smit 2-min (coverage $\geq$ 50%)	L2	Eating	1,083	0.909	0.060	0.000
dog recordings ($\geq$ 60 s)	K-L1	inactive	61	0.000	0.315	0.268
dog recordings ($\geq$ 60 s)	K-L1	locomotion	61	0.000	0.685	0.732
dog natural 2-min windows	K-L1	inactive	130	0.515	0.388	0.000
dog natural 2-min windows	K-L1	locomotion	130	0.246	0.612	1.000
dog recordings ($\geq$ 60 s)	K-L2	Lying on chest	61	0.000	0.155	0.121
dog recordings ($\geq$ 60 s)	K-L2	Sitting	61	0.230	0.083	0.033
dog recordings ($\geq$ 60 s)	K-L2	Standing	61	0.246	0.077	0.016
dog recordings ($\geq$ 60 s)	K-L2	Walking	61	0.000	0.337	0.317
dog recordings ($\geq$ 60 s)	K-L2	Trotting	61	0.000	0.348	0.324
dog natural 2-min windows	K-L2	Lying on chest	130	0.762	0.196	0.000
dog natural 2-min windows	K-L2	Sitting	130	0.877	0.100	0.000
dog natural 2-min windows	K-L2	Standing	130	0.854	0.092	0.000
dog natural 2-min windows	K-L2	Walking	130	0.577	0.296	0.000
dog natural 2-min windows	K-L2	Trotting	130	0.631	0.316	0.000

(b) Share of evaluation samples whose true class proportion lay outside the range of the training samples.

scenario	sample type	variant	class	n outside training range	n evaluated	share outside training range
K-L1	natural 2-min	DR-APP	inactive	0	130	0.000
K-L1	natural 2-min	DR-APP	locomotion	0	130	0.000
K-L1	natural 2-min	DR-natural	inactive	0	130	0.000
K-L1	natural 2-min	DR-natural	locomotion	0	130	0.000
K-L1	recording	DR-APP	inactive	0	61	0.000
K-L1	recording	DR-APP	locomotion	0	61	0.000
K-L1	recording	DR-natural	inactive	0	61	0.000
K-L1	recording	DR-natural	locomotion	0	61	0.000
K-L2	natural 2-min	DR-APP	Lying on chest	19	130	0.146
K-L2	natural 2-min	DR-APP	Sitting	10	130	0.077
K-L2	natural 2-min	DR-APP	Standing	10	130	0.077

continued

scenario	sample type	variant	class	n outside training range	n evaluated	share outside training range
K-L2	natural 2-min	DR-APP	Trotting	32	130	0.246
K-L2	natural 2-min	DR-APP	Walking	32	130	0.246
K-L2	natural 2-min	DR-natural	Lying on chest	0	130	0.000
K-L2	natural 2-min	DR-natural	Sitting	0	130	0.000
K-L2	natural 2-min	DR-natural	Standing	0	130	0.000
K-L2	natural 2-min	DR-natural	Trotting	0	130	0.000
K-L2	natural 2-min	DR-natural	Walking	0	130	0.000
K-L2	recording	DR-APP	Lying on chest	0	61	0.000
K-L2	recording	DR-APP	Sitting	0	61	0.000
K-L2	recording	DR-APP	Standing	0	61	0.000
K-L2	recording	DR-APP	Trotting	0	61	0.000
K-L2	recording	DR-APP	Walking	0	61	0.000
K-L2	recording	DR-natural	Lying on chest	0	61	0.000
K-L2	recording	DR-natural	Sitting	0	61	0.000
K-L2	recording	DR-natural	Standing	0	61	0.000
K-L2	recording	DR-natural	Trotting	0	61	0.000
K-L2	recording	DR-natural	Walking	0	61	0.000
S1	2-min	DR-APP	maintenance	39	1,083	0.036
S1	2-min	DR-APP	inactive	642	1,083	0.593
S1	2-min	DR-APP	locomotion	0	1,083	0.000
S1	2-min	DR-natural	maintenance	0	1,083	0.000
S1	2-min	DR-natural	inactive	0	1,083	0.000
S1	2-min	DR-natural	locomotion	1	1,083	0.001
S1	30-min	DR-APP	maintenance	0	68	0.000
S1	30-min	DR-APP	inactive	20	68	0.294
S1	30-min	DR-APP	locomotion	0	68	0.000
S1	30-min	DR-natural	maintenance	0	68	0.000
S1	30-min	DR-natural	inactive	0	68	0.000
S1	30-min	DR-natural	locomotion	0	68	0.000
S1	APP 120-s	DR-APP	maintenance	1	600	0.002
S1	APP 120-s	DR-APP	inactive	0	600	0.000
S1	APP 120-s	DR-APP	locomotion	0	600	0.000
S1	APP 120-s	DR-natural	maintenance	0	600	0.000
S1	APP 120-s	DR-natural	inactive	0	600	0.000
S1	APP 120-s	DR-natural	locomotion	58	600	0.097
S1-L2 (collar)	30-min	DR-APP	Eating	0	68	0.000
S1-L2 (collar)	30-min	DR-APP	Grooming	0	68	0.000
S1-L2 (collar)	30-min	DR-APP	Lying	19	68	0.279
S1-L2 (collar)	30-min	DR-APP	Sitting	6	68	0.088
S1-L2 (collar)	30-min	DR-APP	Standing	0	68	0.000
S1-L2 (collar)	30-min	DR-APP	Trotting	0	68	0.000
S1-L2 (collar)	30-min	DR-APP	Walking	0	68	0.000
S1-L2 (collar)	30-min	DR-natural	Eating	0	68	0.000
S1-L2 (collar)	30-min	DR-natural	Grooming	0	68	0.000
S1-L2 (collar)	30-min	DR-natural	Lying	0	68	0.000
S1-L2 (collar)	30-min	DR-natural	Sitting	0	68	0.000
S1-L2 (collar)	30-min	DR-natural	Standing	0	68	0.000
S1-L2 (collar)	30-min	DR-natural	Trotting	0	68	0.000
S1-L2 (collar)	30-min	DR-natural	Walking	0	68	0.000
S2	30-min	DR-APP	maintenance	0	68	0.000
S2	30-min	DR-APP	inactive	20	68	0.294
S2	30-min	DR-APP	locomotion	0	68	0.000
S2	30-min	DR-natural	maintenance	0	68	0.000
S2	30-min	DR-natural	inactive	0	68	0.000
S2	30-min	DR-natural	locomotion	0	68	0.000
S3 [exploratory]	cat sample	DR-APP	maintenance	0	8	0.000
S3 [exploratory]	cat sample	DR-APP	inactive	2	8	0.250

continued

scenario	sample type	variant	class	n outside training range	n evaluated	share outside training range
S3 [exploratory]	cat sample	DR-APP	locomotion	0	8	0.000
S3 [exploratory]	cat sample	DR-natural	maintenance	0	8	0.000
S3 [exploratory]	cat sample	DR-natural	inactive	0	8	0.000
S3 [exploratory]	cat sample	DR-natural	locomotion	1	8	0.125
S4-Dunford-closed [exploratory]	cat sample	DR-APP	inactive	0	7	0.000
S4-Dunford-closed [exploratory]	cat sample	DR-APP	locomotion	0	7	0.000
S4-Dunford-closed [exploratory]	cat sample	DR-natural	inactive	0	7	0.000
S4-Dunford-closed [exploratory]	cat sample	DR-natural	locomotion	0	7	0.000
S4-Smit-closed	30-min	DR-APP	inactive	0	68	0.000
S4-Smit-closed	30-min	DR-APP	locomotion	0	68	0.000
S4-Smit-closed	30-min	DR-natural	inactive	0	68	0.000
S4-Smit-closed	30-min	DR-natural	locomotion	0	68	0.000

(c) Selected number of PLS components and number of training samples per outer fold.

scenario	variant	descriptor	components median	components min	components max	n training samples median	n training samples min	n training samples max	n outer folds
K-L1	DR-APP	posterior histograms (h)	2.000	2	3	3,200	3,200	3,200	5
K-L1	DR-APP	feature deciles (q)	4.000	2	4	3,200	3,200	3,200	5
K-L1	DR-natural	posterior histograms (h)	2.000	1	12	152	150	156	5
K-L1	DR-natural	feature deciles (q)	6.000	6	8	152	150	156	5
K-L2	DR-APP	posterior histograms (h)	3.000	2	3	2,569	2,446	2,629	5
K-L2	DR-APP	feature deciles (q)	3.000	1	5	2,569	2,446	2,629	5
K-L2	DR-natural	posterior histograms (h)	5.000	4	10	152	151	156	5
K-L2	DR-natural	feature deciles (q)	9.000	9	13	152	151	156	5
S1	DR-APP	posterior histograms (h)	2.500	2	17	1,100	1,100	1,100	12
S1	DR-APP	feature deciles (q)	4.000	2	16	1,100	1,100	1,100	12
S1	DR-natural	posterior histograms (h)	3.000	2	9	1,046	1,035	1,113	12
S1	DR-natural	feature deciles (q)	14.500	13	17	1,046	1,035	1,113	12
S1-L2 (collar)	DR-APP	posterior histograms (h)	2.000	1	5	1,037	1,037	1,100	12
S1-L2 (collar)	DR-APP	feature deciles (q)	1.000	1	8	1,037	1,037	1,100	12
S1-L2 (collar)	DR-natural	posterior histograms (h)	8.000	4	15	1,046	1,035	1,113	12
S1-L2 (collar)	DR-natural	feature deciles (q)	20	19	20	1,046	1,035	1,113	12
S2	DR-APP	posterior histograms (h)	2.000	2	3	1,100	1,100	1,100	12
S2	DR-APP	feature deciles (q)	4.000	2	4	1,100	1,100	1,100	12
S2	DR-natural	posterior histograms (h)	5.000	3	19	1,046	1,035	1,113	12
S2	DR-natural	feature deciles (q)	14	12	20	1,046	1,035	1,113	12
S3 [exploratory]	DR-APP	posterior histograms (h)	20	20	20	1,200	1,200	1,200	1
S3 [exploratory]	DR-APP	feature deciles (q)	5.000	5	5	1,200	1,200	1,200	1
S3 [exploratory]	DR-natural	posterior histograms (h)	15	15	15	1,151	1,151	1,151	1
S3 [exploratory]	DR-natural	feature deciles (q)	15	15	15	1,151	1,151	1,151	1
S4-Dunford-closed [exploratory]	DR-APP	posterior histograms (h)	4.000	4	4	4,000	4,000	4,000	1
S4-Dunford-closed [exploratory]	DR-APP	feature deciles (q)	3.000	3	3	4,000	4,000	4,000	1
S4-Dunford-closed [exploratory]	DR-natural	posterior histograms (h)	1.000	1	1	191	191	191	1
S4-Dunford-closed [exploratory]	DR-natural	feature deciles (q)	6.000	6	6	191	191	191	1
S4-Smit-closed	DR-APP	posterior histograms (h)	4.000	4	4	4,000	4,000	4,000	1
S4-Smit-closed	DR-APP	feature deciles (q)	3.000	3	3	4,000	4,000	4,000	1
S4-Smit-closed	DR-natural	posterior histograms (h)	1.000	1	1	191	191	191	1
S4-Smit-closed	DR-natural	feature deciles (q)	6.000	6	6	191	191	191	1

Table S7. Relative absolute error (RAE) of classify-and-count and each estimator, and signed bias of the rarest class.

The first 133 rows are comparisons of group A of Table S3; RAE could be computed for 129 of them (window-level class proportions were not stored for the four S2-L2 (mapped) rows), and 21 have direction_mismatch = TRUE. d_RAE = RAE_CC − RAE_arm (mean over individuals; 95% individual-bootstrap interval). direction_mismatch = TRUE when the RAE interval lies above zero but the AE decision is not 'better' (Table 4 of the paper). Rarest class = class with the smallest mean true proportion; its signed bias is in percentage points and is given for equivalent and inconclusive comparisons. The last rows are target-labelled arms (m = 1,800, random blocks), for which RAE was computed without an interval. 16 rows were computed in round R15 for cells added to Table 3 after the RAE round.

Scenario	Samples	Arm	AE decision (d)	n	RAE CC	RAE arm	d_{RAE} [95% CI]	Mis-match	Rarest class	True %	Bias CC (pp)	Bias arm (pp)	Note
S0 (leaky)	Smit 30-min (68)	PCC	inconclusive (−0.010)	12	1.566	2.643	−1.177 [−1.635, −0.812]		locomotion	3.1	−1.3	−0.0	
S0 (leaky)	Smit 30-min (68)	ACC	equivalent (−0.003)	12	1.566	0.636	+1.161 [+0.359, +2.056]	yes	locomotion	3.1	−1.3	+0.7	
S0 (leaky)	Smit 30-min (68)	PACC	worse (−0.012)	12	1.566	0.729	+1.058 [+0.206, +2.022]	yes					
S0 (leaky)	Smit 30-min (68)	EMQ	worse (−0.021)	12	1.566	0.844	+0.992 [+0.008, +2.092]	yes					
S1	Smit 30-min (68)	N-target	worse (−0.083)	12	3.986	38.191	−38.958 [−73.285, −15.680]						
S1	Smit 30-min (68)	PCC	worse (−0.015)	12	3.986	7.950	−4.144 [−6.180, −2.571]						
S1	Smit 30-min (68)	ACC	inconclusive (−0.006)	12	3.986	0.954	+3.500 [+0.691, +7.383]	yes	locomotion	3.2	−0.9	+0.9	
S1	Smit 30-min (68)	PACC	worse (−0.024)	12	3.986	2.003	+2.546 [−0.895, +7.081]						
S1	Smit 30-min (68)	EMQ	worse (−0.044)	12	3.986	3.712	+1.233 [−4.129, +6.448]						
S1	Smit 30-min (68)	DM	worse (−0.033)	12	3.986	2.485	+2.303 [−1.517, +6.251]						
S1	Smit 30-min (68)	DR-APP	worse (−0.127)	12	3.986	61.950	−64.550 [−145.936, −18.177]						
S1	Smit 30-min (68)	DR-natural	worse (−0.051)	12	3.986	2.566	+1.651 [−0.102, +4.049]						
S2	Smit 30-min (68)	N-target	worse (−0.067)	12	10.023	38.191	−32.350 [−64.925, −10.379]						
S2	Smit 30-min (68)	PCC	worse (−0.031)	12	10.023	18.493	−8.594 [−12.987, −4.829]						
S2	Smit 30-min (68)	ACC	worse (−0.012)	12	10.023	5.044	+5.699 [+1.114, +11.663]	yes					
S2	Smit 30-min (68)	PACC	worse (−0.069)	12	10.023	12.902	−1.441 [−13.235, +10.357]						
S2	Smit 30-min (68)	EMQ	worse (−0.134)	12	10.023	26.889	−14.412 [−43.119, +7.845]						
S2	Smit 30-min (68)	DM	worse (−0.098)	12	10.023	11.349	+0.101 [−10.899, +11.215]						
S2	Smit 30-min (68)	DR-APP	worse (−0.186)	12	10.023	68.897	−66.431 [−130.883, −25.266]						
S2	Smit 30-min (68)	DR-natural	worse (−0.041)	12	10.023	2.744	+8.040 [+3.772, +13.005]	yes					
S3 [exploratory]	Dunford cat sample	N-target	inconclusive (−0.083)	8	12.404	27.420	−15.016 [−49.136, +7.648]		maintenance	7.9	+12.0	−1.0	
S3 [exploratory]	Dunford cat sample	PCC	inconclusive (−0.032)	8	12.404	20.760	−8.356 [−16.378, −1.033]		maintenance	7.9	+12.0	+16.1	
S3 [exploratory]	Dunford cat sample	ACC	inconclusive (+0.039)	8	12.404	6.519	+5.884 [+1.586, +10.590]	yes	maintenance	7.9	+12.0	+4.0	
S3 [exploratory]	Dunford cat sample	PACC	inconclusive (+0.014)	8	12.404	14.526	−2.122 [−8.565, +5.028]		maintenance	7.9	+12.0	+11.4	
S3 [exploratory]	Dunford cat sample	EMQ	inconclusive (−0.048)	8	12.404	14.377	−1.973 [−13.207, +8.298]		maintenance	7.9	+12.0	+24.7	
S4-Smit-open	Smit 30-min (68)	N-target	inconclusive (−0.021)	12	4.716	38.191	−38.652 [−77.894, −12.558]		locomotion	3.2	+3.6	+0.0	
S4-Smit-open	Smit 30-min (68)	PCC	equivalent (−0.000)	12	4.716	5.552	−0.850 [−1.272, −0.480]		locomotion	3.2	+3.6	+4.1	
S4-Smit-open	Smit 30-min (68)	ACC	equivalent (−0.001)	12	4.716	2.398	+2.524 [+1.170, +4.045]	yes	locomotion	3.2	+3.6	+2.2	
S4-Smit-open	Smit 30-min (68)	PACC	equivalent (−0.001)	12	4.716	2.669	+2.268 [+1.025, +3.701]	yes	locomotion	3.2	+3.6	+2.5	
S4-Smit-open	Smit 30-min (68)	EMQ	equivalent (−0.006)	12	4.716	1.070	+3.818 [+1.746, +6.213]	yes	locomotion	3.2	+3.6	−0.7	
S4-Smit-closed	Smit 30-min (68)	N-target	worse (−0.033)	12	1.359	17.933	−17.720 [−26.611, −10.290]						
S4-Smit-closed	Smit 30-min (68)	PCC	equivalent (−0.001)	12	1.359	1.825	−0.502 [−0.760, −0.266]		locomotion	4.1	+0.7	+0.9	
S4-Smit-closed	Smit 30-min (68)	ACC	equivalent (−0.001)	12	1.359	0.180	+1.154 [+0.489, +1.898]	yes	locomotion	4.1	+0.7	−0.5	
S4-Smit-closed	Smit 30-min (68)	PACC	equivalent (−0.002)	12	1.359	0.187	+1.147 [+0.481, +1.892]	yes	locomotion	4.1	+0.7	−0.5	
S4-Smit-closed	Smit 30-min (68)	EMQ	worse (−0.010)	12	1.359	0.265	+1.077 [+0.409, +1.817]	yes					
S4-Dunford-open [exploratory]	Dunford cat sample	N-target	inconclusive (−0.123)	8	0.492	27.420	−26.928 [−62.990, −2.896]		maintenance	7.9	−7.9	−1.0	
S4-Dunford-open [exploratory]	Dunford cat sample	PCC	equivalent (−0.004)	8	0.492	0.620	−0.127 [−0.308, −0.015]		maintenance	7.9	−7.9	−7.9	
S4-Dunford-open [exploratory]	Dunford cat sample	ACC	inconclusive (−0.007)	8	0.492	0.535	−0.042 [−0.072, −0.018]		maintenance	7.9	−7.9	−7.9	
S4-Dunford-open [exploratory]	Dunford cat sample	PACC	inconclusive (−0.010)	8	0.492	0.552	−0.059 [−0.092, −0.030]		maintenance	7.9	−7.9	−7.9	

continued

Scenario	Samples	Arm	AE decision (d)	n	RAE CC	RAE arm	d_{RAE} [95% CI]	Mis-match	Rarest class	True %	Bias CC (pp)	Bias arm (pp)	Note
S4-Dunford-open [exploratory]	Dunford cat sample	EMQ	inconclusive (−0.026)	8	0.492	0.539	−0.046 [−0.135, +0.084]		maintenance	7.9	−7.9	−7.9	
S4-Dunford-closed [exploratory]	Dunford cat sample	N-target	inconclusive (−0.207)	7	0.440	31.804	−31.364 [−81.677, −0.710]		locomotion	30.9	−8.6	+0.8	
S4-Dunford-closed [exploratory]	Dunford cat sample	PCC	inconclusive (−0.007)	7	0.440	0.660	−0.220 [−0.515, −0.031]		locomotion	30.9	−8.6	−9.2	
S4-Dunford-closed [exploratory]	Dunford cat sample	ACC	inconclusive (−0.010)	7	0.440	0.500	−0.060 [−0.112, −0.020]		locomotion	30.9	−8.6	−9.6	
S4-Dunford-closed [exploratory]	Dunford cat sample	PACC	inconclusive (−0.015)	7	0.440	0.529	−0.089 [−0.141, −0.042]		locomotion	30.9	−8.6	−10.2	
S4-Dunford-closed [exploratory]	Dunford cat sample	EMQ	inconclusive (−0.036)	7	0.440	0.481	−0.042 [−0.200, +0.169]		locomotion	30.9	−8.6	−12.3	
R-D [exploratory]	Dunford cat sample	N-target	inconclusive (−0.158)	8	0.790	27.420	−26.630 [−62.174, −2.871]		maintenance	7.9	−7.8	−1.0	
R-D [exploratory]	Dunford cat sample	PCC	inconclusive (−0.007)	8	0.790	5.469	−4.679 [−8.813, −0.982]		maintenance	7.9	−7.8	−4.6	
R-D [exploratory]	Dunford cat sample	ACC	inconclusive (−0.112)	8	0.790	4.543	−3.753 [−11.441, +0.810]		maintenance	7.9	−7.8	+11.3	
R-D [exploratory]	Dunford cat sample	PACC	inconclusive (−0.071)	8	0.790	4.006	−3.216 [−10.443, +0.843]		maintenance	7.9	−7.8	+3.8	
R-D [exploratory]	Dunford cat sample	EMQ	inconclusive (−0.054)	8	0.790	7.355	−6.565 [−16.482, +0.570]		maintenance	7.9	−7.8	+1.6	
S1-closed	Smit 30-min (68)	PCC	inconclusive (−0.008)	12	0.557	3.756	−2.966 [−4.278, −1.683]		locomotion	4.1	−1.0	+1.4	
S1-closed	Smit 30-min (68)	ACC	inconclusive (−0.011)	12	0.557	0.210	+0.314 [+0.031, +0.611]	yes	locomotion	4.1	−1.0	+1.5	
S1-closed	Smit 30-min (68)	PACC	inconclusive (−0.041)	12	0.557	1.938	−1.091 [−3.956, +0.524]		locomotion	4.1	−1.0	+4.0	
S1-closed	Smit 30-min (68)	EMQ	worse (−0.080)	12	0.557	4.365	−3.051 [−9.780, +0.493]						
S1-L2 (collar)	Smit 30-min (68)	PCC	equivalent (−0.001)	12	52.409	57.793	−4.169 [−11.087, +2.829]		Grooming	0.3	−0.5	−0.2	
S1-L2 (collar)	Smit 30-min (68)	ACC	worse (−0.020)	12	52.409	52.981	−3.161 [−16.514, +9.360]						
S1-L2 (collar)	Smit 30-min (68)	PACC	worse (−0.024)	12	52.409	56.821	−7.149 [−21.649, +6.843]						
S1-L2 (collar)	Smit 30-min (68)	EMQ	worse (−0.020)	12	52.409	48.905	+2.673 [−9.422, +12.036]						
S1-L2 (collar)	Smit 30-min (68)	DM	inconclusive (−0.013)	12	52.410	53.398	−3.040 [−22.971, +14.609]		Trotting	0.3	−0.5	+0.6	
S1-L2 (collar)	Smit 30-min (68)	DR-APP	inconclusive (−0.029)	12	52.410	82.813	−21.381 [−61.482, +31.918]		Trotting	0.3	−0.5	+0.8	
S1-L2 (collar)	Smit 30-min (68)	DR-natural	inconclusive (−0.002)	12	52.410	45.158	+15.188 [−15.225, +56.128]		Trotting	0.3	−0.5	+0.4	
S1-L2 (harness)	Smit 30-min (68)	PCC	worse (−0.014)	12	8.208	18.794	−11.036 [−14.676, −7.701]						
S1-L2 (harness)	Smit 30-min (68)	ACC	worse (−0.015)	12	8.208	5.234	+3.263 [+1.001, +5.534]	yes					
S1-L2 (harness)	Smit 30-min (68)	PACC	worse (−0.022)	12	8.208	7.152	+1.680 [−1.536, +4.923]						
S1-L2 (harness)	Smit 30-min (68)	EMQ	worse (−0.022)	12	8.208	10.389	−1.532 [−9.489, +4.072]						
S2-L2 (mapped)	Smit 30-min (68)	PCC	inconclusive (−0.003)										S2-L2 (mapped): window-level class proportions not stored (AE only)
S2-L2 (mapped)	Smit 30-min (68)	ACC	worse (−0.013)										S2-L2 (mapped): window-level class proportions not stored (AE only)
S2-L2 (mapped)	Smit 30-min (68)	PACC	worse (−0.025)										S2-L2 (mapped): window-level class proportions not stored (AE only)
S2-L2 (mapped)	Smit 30-min (68)	EMQ	worse (−0.058)										S2-L2 (mapped): window-level class proportions not stored (AE only)
S1 natural 2-min	Smit 2-min	ACC	equivalent (−0.001)	12	2.129	1.194	+1.000 [+0.748, +1.284]	yes					
S1 natural 2-min	Smit 2-min	PACC	inconclusive (−0.018)	12	2.129	2.341	−0.095 [−1.692, +0.904]						
S1 natural 2-min	Smit 2-min	EMQ	worse (−0.050)	12	2.129	4.117	−1.760 [−6.300, +0.654]						
S1 natural 2-min	Smit 2-min	DM	inconclusive (−0.028)	12	2.129	2.496	−0.189 [−2.638, +1.253]		locomotion	3.5	−1.1	+6.5	
S1 natural 2-min	Smit 2-min	DR-APP	worse (−0.232)	12	2.129	25.368	−23.401 [−28.180, −20.064]						

continued

Scenario	Samples	Arm	AE decision (d)	n	RAE CC	RAE arm	d_{RAE} [95% CI]	Mis-match	Rarest class	True %	Bias CC (pp)	Bias arm (pp)	Note
S1 natural 2-min	Smit 2-min	DR-natural	worse (−0.021)	12	2.129	1.891	+0.078 [−0.697, +0.738]						
S1 APP 120-s (bridge)	APP 120-s	ACC	better (+0.061)	12	1.367	0.666	+0.700 [+0.511, +0.896]						
S1 APP 120-s (bridge)	APP 120-s	PACC	better (+0.060)	12	1.367	0.679	+0.688 [+0.517, +0.856]						
S1 APP 120-s (bridge)	APP 120-s	EMQ	better (+0.050)	12	1.367	0.751	+0.615 [+0.356, +0.837]						
S1 APP 120-s (bridge)	APP 120-s	DM	better (+0.048)	12	1.367	0.727	+0.639 [+0.469, +0.811]						
S1 APP 120-s (bridge)	APP 120-s	DR-APP	better (+0.034)	12	1.367	0.858	+0.509 [+0.276, +0.734]						
S1 APP 120-s (bridge)	APP 120-s	DR-natural	worse (−0.107)	12	1.367	1.321	+0.046 [−0.155, +0.263]						
K-L1 within-dog	dog recordings	PCC	inconclusive (−0.008)	44	0.045	0.091	−0.054 [−0.109, −0.015]						computed from stored posteriors (round R11)
K-L1 within-dog	dog recordings	ACC	equivalent (−0.003)	44	0.045	0.075	−0.031 [−0.048, −0.017]						computed from stored posteriors (round R11)
K-L1 within-dog	dog recordings	PACC	worse (−0.011)	44	0.045	0.128	−0.089 [−0.141, −0.049]						computed from stored posteriors (round R11)
K-L1 within-dog	dog recordings	EMQ	equivalent (−0.006)	44	0.045	0.073	−0.031 [−0.058, −0.005]						computed from stored posteriors (round R11)
K-L1 within-dog	dog recordings	DM	equivalent (−0.005)	44	0.045	0.063	−0.021 [−0.049, +0.007]		inactive	31.5	−0.5	−0.7	
K-L1 within-dog	dog recordings	DR-APP	worse (−0.066)	44	0.045	0.259	−0.212 [−0.269, −0.156]						
K-L1 within-dog	dog recordings	DR-natural	worse (−0.122)	44	0.045	0.558	−0.517 [−0.745, −0.327]						
K-L1 within-dog	dog 2-min	PCC	worse (−0.015)	35	0.568	1.329	−1.159 [−1.853, −0.592]						computed from stored posteriors (round R11)
K-L1 within-dog	dog 2-min	ACC	equivalent (−0.001)	35	0.568	0.526	+0.072 [+0.021, +0.131]	yes					computed from stored posteriors (round R11)
K-L1 within-dog	dog 2-min	PACC	inconclusive (−0.010)	35	0.568	0.930	−0.644 [−1.197, −0.203]						computed from stored posteriors (round R11)
K-L1 within-dog	dog 2-min	EMQ	inconclusive (+0.009)	35	0.568	0.082	+0.784 [+0.246, +1.454]	yes					computed from stored posteriors (round R11)
K-L1 within-dog	dog 2-min	DM	inconclusive (+0.009)	35	0.568	0.042	+0.879 [+0.235, +1.819]	yes	inactive	38.8	+0.5	−0.0	
K-L1 within-dog	dog 2-min	DR-APP	worse (−0.113)	35	0.568	7.512	−7.519 [−9.450, −5.805]						
K-L1 within-dog	dog 2-min	DR-natural	worse (−0.072)	35	0.568	3.633	−4.239 [−6.666, −2.415]						
K-L1 within-dog	APP 120-s	PCC	inconclusive (−0.007)	40	0.079	0.158	−0.079 [−0.140, −0.033]						computed from stored posteriors (round R11)
K-L1 within-dog	APP 120-s	ACC	equivalent (−0.003)	40	0.079	0.105	−0.026 [−0.040, −0.013]						computed from stored posteriors (round R11)
K-L1 within-dog	APP 120-s	PACC	inconclusive (−0.010)	40	0.079	0.153	−0.074 [−0.102, −0.049]						computed from stored posteriors (round R11)
K-L1 within-dog	APP 120-s	EMQ	equivalent (−0.003)	40	0.079	0.071	+0.007 [−0.030, +0.052]						computed from stored posteriors (round R11)
K-L2 within-dog	dog recordings	PCC	equivalent (−0.002)	44	3.354	3.563	−0.287 [−1.033, +0.541]						computed from stored posteriors (round R11)

continued

Scenario	Samples	Arm	AE decision (d)	n	RAE CC	RAE arm	d_{RAE} [95% CI]	Mis-match	Rarest class	True %	Bias CC (pp)	Bias arm (pp)	Note
K-L2 within-dog	dog recordings	ACC	worse (−0.031)	44	3.354	2.968	−0.155 [−1.834, +1.751]						computed from stored posteriors (round R11)
K-L2 within-dog	dog recordings	PACC	worse (−0.031)	44	3.354	3.351	−0.671 [−2.498, +1.454]						computed from stored posteriors (round R11)
K-L2 within-dog	dog recordings	EMQ	worse (−0.021)	44	3.354	4.456	−0.911 [−2.022, +0.016]						computed from stored posteriors (round R11)
K-L2 within-dog	dog recordings	DM	worse (−0.029)	44	3.354	1.907	+1.013 [−0.299, +2.801]						
K-L2 within-dog	dog recordings	DR-APP	worse (−0.045)	44	3.354	3.445	−0.088 [−1.662, +1.767]						
K-L2 within-dog	dog recordings	DR-natural	worse (−0.058)	44	3.354	2.096	+1.038 [−0.775, +3.327]						
K-L2 within-dog	dog 2-min	PCC	inconclusive (−0.011)	35	7.268	8.540	−0.836 [−1.539, −0.107]						computed from stored posteriors (round R11)
K-L2 within-dog	dog 2-min	ACC	inconclusive (−0.009)	35	7.268	7.539	−0.434 [−1.501, +0.697]						computed from stored posteriors (round R11)
K-L2 within-dog	dog 2-min	PACC	inconclusive (−0.010)	35	7.268	7.284	−0.508 [−1.588, +0.621]						computed from stored posteriors (round R11)
K-L2 within-dog	dog 2-min	EMQ	equivalent (−0.001)	35	7.268	7.359	−0.058 [−0.717, +0.594]						computed from stored posteriors (round R11)
K-L2 within-dog	dog 2-min	DM	inconclusive (+0.007)	35	7.268	6.045	+0.898 [−0.263, +2.235]		Standing	9.2	−2.8	−2.4	
K-L2 within-dog	dog 2-min	DR-APP	worse (−0.107)	35	7.268	15.792	−8.485 [−10.948, −5.939]						
K-L2 within-dog	dog 2-min	DR-natural	worse (−0.054)	35	7.268	10.802	−4.026 [−5.982, −2.004]						
K-L2 within-dog	APP 120-s	PCC	equivalent (−0.002)	38	2.142	2.267	−0.143 [−0.526, +0.387]						computed from stored posteriors (round R11)
K-L2 within-dog	APP 120-s	ACC	worse (−0.036)	38	2.142	2.438	−0.464 [−1.312, +0.459]						computed from stored posteriors (round R11)
K-L2 within-dog	APP 120-s	PACC	worse (−0.036)	38	2.142	2.344	−0.565 [−1.408, +0.336]						computed from stored posteriors (round R11)
K-L2 within-dog	APP 120-s	EMQ	worse (−0.023)	38	2.142	2.704	−0.562 [−1.316, +0.025]						computed from stored posteriors (round R11)
S3 [exploratory]	Dunford cat sample	DM	inconclusive (+0.012)	8	12.404	6.743	+5.661 [+0.680, +11.577]	yes					computed in round R15
S3 [exploratory]	Dunford cat sample	DR-APP	inconclusive (−0.142)	8	12.404	71.884	−59.480 [−119.856, −0.638]						computed in round R15
S3 [exploratory]	Dunford cat sample	DR-natural	inconclusive (−0.094)	8	12.404	34.700	−22.296 [−43.948, −4.757]						computed in round R15
S4-Smit-closed	Smit 30-min (68)	DM	equivalent (−0.007)	12	1.359	0.226	+1.111 [+0.447, +1.859]	yes					computed in round R15
S4-Smit-closed	Smit 30-min (68)	DR-APP	worse (−0.079)	12	1.359	59.038	−63.456 [−114.051, −31.015]						computed in round R15
S4-Smit-closed	Smit 30-min (68)	DR-natural	worse (−0.024)	12	1.359	2.521	−1.015 [−2.252, −0.224]						computed in round R15
S4-Dunford-closed [exploratory]	Dunford cat sample	DM	inconclusive (−0.023)	7	0.440	0.447	−0.008 [−0.175, +0.266]						computed in round R15
S4-Dunford-closed [exploratory]	Dunford cat sample	DR-APP	inconclusive (−0.112)	7	0.440	41.724	−41.284 [−109.178, +0.308]						computed in round R15
S4-Dunford-closed [exploratory]	Dunford cat sample	DR-natural	inconclusive (−0.140)	7	0.440	38.109	−37.669 [−100.809, +0.319]						computed in round R15
S1-L2 (collar)	Smit 30-min (68)	N-target	worse (−0.036)	12	52.409	86.879	−33.047 [−52.033, −13.721]						computed in round R15

continued

Scenario	Samples	Arm	AE decision (d)	n	RAE CC	RAE arm	d_{RAE} [95% CI]	Mis-match	Rarest class	True %	Bias CC (pp)	Bias arm (pp)	Note
K-L1 within-dog	APP 120-s	N-target	worse (−0.237)	40	0.079	2.297	−2.219 [−2.863, −1.676]						computed in round R15
K-L1 within-dog	dog recordings	N-target	worse (−0.200)	44	0.045	1.334	−1.321 [−1.857, −0.861]						computed in round R15
K-L1 within-dog	dog 2-min	N-target	worse (−0.407)	35	0.568	30.039	−29.308 [−33.543, −25.405]						computed in round R15
K-L2 within-dog	APP 120-s	N-target	worse (−0.079)	38	2.142	3.261	−1.323 [−2.452, +0.035]						computed in round R15
K-L2 within-dog	dog recordings	N-target	worse (−0.063)	44	3.354	5.085	−1.860 [−3.837, +0.373]						computed in round R15
K-L2 within-dog	dog 2-min	N-target	worse (−0.195)	35	7.268	23.216	−15.399 [−17.820, −12.445]						computed in round R15
S1		Retraining (m = 1800, random blocks)	equivalent (+0.002)	12	4.135	4.014			locomotion	3.7	−1.1	−1.1	target-labelled arm (m = 1,800, random blocks); interval not computed
S1		Weighted retraining (m = 1800, random blocks)	equivalent (+0.002)	12	4.135	4.150			locomotion	3.7	−1.1	−1.1	target-labelled arm (m = 1,800, random blocks); interval not computed
S2		Retraining (m = 1800, random blocks)	inconclusive (+0.007)	12	9.251	7.428			locomotion	3.7	+1.9	+0.9	target-labelled arm (m = 1,800, random blocks); interval not computed
S2		Weighted retraining (m = 1800, random blocks)	inconclusive (+0.007)	12	9.251	7.309			locomotion	3.7	+1.9	+0.8	target-labelled arm (m = 1,800, random blocks); interval not computed
S3 [exploratory]		T-ACC (m = 1800, random blocks)	inconclusive (+0.003)	8	12.404	6.649			maintenance	7.9	+12.0	+10.1	target-labelled arm (m = 1,800, random blocks); interval not computed
S3 [exploratory]		PACC-target (m = 1800, random blocks)	inconclusive (+0.022)	8	12.404	7.360			maintenance	7.9	+12.0	+8.6	target-labelled arm (m = 1,800, random blocks); interval not computed
S3 [exploratory]		PACC-shrinkage (m = 1800, random blocks)	inconclusive (+0.022)	8	12.404	7.742			maintenance	7.9	+12.0	+7.9	target-labelled arm (m = 1,800, random blocks); interval not computed
S3 [exploratory]		Retraining (m = 1800, random blocks)	inconclusive (+0.004)	8	12.404	12.194			maintenance	7.9	+12.0	+11.1	target-labelled arm (m = 1,800, random blocks); interval not computed
S3 [exploratory]		Weighted retraining (m = 1800, random blocks)	inconclusive (−0.002)	8	12.404	12.048			maintenance	7.9	+12.0	+11.4	target-labelled arm (m = 1,800, random blocks); interval not computed

continued

Scenario	Samples	Arm	AE decision (d)	n	RAE CC	RAE arm	d_{RAE} [95% CI]	Mis-match	Rarest class	True %	Bias CC (pp)	Bias arm (pp)	Note
S3 [exploratory]		Target-only, paired (m = 1800, random blocks)	inconclusive (+0.076)	8	12.404	0.782			maintenance	7.9	+12.0	−7.9	target-labelled arm (m = 1,800, random blocks); interval not computed
S4-Smit-closed		Retraining (m = 1800, random blocks)	equivalent (+0.000)	12	1.330	1.303			locomotion	4.9	+0.9	+0.8	target-labelled arm (m = 1,800, random blocks); interval not computed
S4-Dunford-open [exploratory]		T-ACC (m = 1800, random blocks)	inconclusive (+0.011)	8	0.492	0.398			maintenance	7.9	−7.9	−7.9	target-labelled arm (m = 1,800, random blocks); interval not computed
S4-Dunford-open [exploratory]		PACC-target (m = 1800, random blocks)	inconclusive (+0.012)	8	0.492	0.396			maintenance	7.9	−7.9	−7.9	target-labelled arm (m = 1,800, random blocks); interval not computed
S4-Dunford-open [exploratory]		PACC-shrinkage (m = 1800, random blocks)	inconclusive (+0.016)	8	0.492	0.389			maintenance	7.9	−7.9	−7.9	target-labelled arm (m = 1,800, random blocks); interval not computed
S4-Dunford-open [exploratory]		Retraining (m = 1800, random blocks)	inconclusive (+0.021)	8	0.492	0.694			maintenance	7.9	−7.9	−7.9	target-labelled arm (m = 1,800, random blocks); interval not computed
S4-Dunford-open [exploratory]		Weighted retraining (m = 1800, random blocks)	inconclusive (+0.021)	8	0.492	0.655			maintenance	7.9	−7.9	−7.9	target-labelled arm (m = 1,800, random blocks); interval not computed
S4-Dunford-open [exploratory]		Target-only, paired (m = 1800, random blocks)	inconclusive (+0.036)	8	0.492	0.827			maintenance	7.9	−7.9	−7.9	target-labelled arm (m = 1,800, random blocks); interval not computed
S4-Dunford-closed [exploratory]		T-ACC (m = 1800, random blocks)	inconclusive (+0.005)	7	0.440	0.312			locomotion	30.9	−8.6	+1.1	target-labelled arm (m = 1,800, random blocks); interval not computed
S4-Dunford-closed [exploratory]		PACC-target (m = 1800, random blocks)	inconclusive (+0.008)	7	0.440	0.305			locomotion	30.9	−8.6	+1.1	target-labelled arm (m = 1,800, random blocks); interval not computed
S4-Dunford-closed [exploratory]		PACC-shrinkage (m = 1800, random blocks)	inconclusive (+0.015)	7	0.440	0.295			locomotion	30.9	−8.6	−0.7	target-labelled arm (m = 1,800, random blocks); interval not computed

continued

Scenario	Samples	Arm	AE decision (d)	n	RAE CC	RAE arm	d_{RAE} [95% CI]	Mis-match	Rarest class	True %	Bias CC (pp)	Bias arm (pp)	Note
S4-Dunford-closed [exploratory]		Retraining (m = 1800, random blocks)	inconclusive (+0.028)	7	0.440	0.820			locomotion	30.9	−8.6	−5.6	target-labelled arm (m = 1,800, random blocks); interval not computed
S4-Dunford-closed [exploratory]		Weighted retraining (m = 1800, random blocks)	inconclusive (+0.030)	7	0.440	0.695			locomotion	30.9	−8.6	−5.4	target-labelled arm (m = 1,800, random blocks); interval not computed
S4-Dunford-closed [exploratory]		Target-only, paired (m = 1800, random blocks)	inconclusive (+0.056)	7	0.440	0.924			locomotion	30.9	−8.6	−2.6	target-labelled arm (m = 1,800, random blocks); interval not computed
S2-L2 (mapped)		PCC	inconclusive (−0.003)	12									target-labelled arm (m = 1,800, random blocks); interval not computed
S4-Smit-closed 30-min		DM	equivalent (−0.007)	12	1.359	0.226			locomotion	4.1	+0.7	−1.3	target-labelled arm (m = 1,800, random blocks); interval not computed
S3 [exploratory] cat sample		DM	inconclusive (+0.012)	8	12.404	6.743			maintenance	7.9	+12.0	+2.7	target-labelled arm (m = 1,800, random blocks); interval not computed
S3 [exploratory] cat sample		DR-APP	inconclusive (−0.142)	8	12.404	71.884			maintenance	7.9	+12.0	+35.0	target-labelled arm (m = 1,800, random blocks); interval not computed
S3 [exploratory] cat sample		DR-natural	inconclusive (−0.094)	8	12.404	34.700			maintenance	7.9	+12.0	+20.7	target-labelled arm (m = 1,800, random blocks); interval not computed
S4-Dunford-closed [exploratory] cat sample		DM	inconclusive (−0.023)	7	0.440	0.447			locomotion	30.9	−8.6	−11.0	target-labelled arm (m = 1,800, random blocks); interval not computed
S4-Dunford-closed [exploratory] cat sample		DR-APP	inconclusive (−0.112)	7	0.440	41.724			locomotion	30.9	−8.6	+18.1	target-labelled arm (m = 1,800, random blocks); interval not computed
S4-Dunford-closed [exploratory] cat sample		DR-natural	inconclusive (−0.140)	7	0.440	38.109			locomotion	30.9	−8.6	+22.6	target-labelled arm (m = 1,800, random blocks); interval not computed

Table S8. Dog-to-cat transfer with maintenance absent from the source (S4-Smit-open): signed bias of the maintenance share under classify-and-count, by averaging level.

Percentage points; 95% individual-bootstrap interval. The paper reports measure (i).

measure	bias pp	ci low	ci high	n individuals	n windows
cat level (all seconds of each cat)	−15.1			12	
(i) individual mean (mean over each cat's 30-min windows, then over cats)	−12.9	−17.390	−8.124	12	68
(ii) window mean (68 windows)	−14.7	−18.467	−10.079	12	68

Table S9. Prevalence shift and shift-dependent decision rules.

(a) Prevalence shift $\delta = \frac{1}{2}\|\mathrm{p} - \pi_\mathrm{source}\|_1$ by sample set: median and 90th percentile.

sample set	n	median delta	p90 delta
S1	68	0.143	0.251
S1 natural 2-min	1,083	0.187	0.621
S1 APP 120-s	600	0.543	0.775
S2	68	0.143	0.251
S3 [exploratory]	8	0.275	0.605
S4-Dunford-closed [exploratory]	7	0.390	0.609
S4-Smit-closed	68	0.594	0.611
K-L1 within-dog APP 120-s	2,000	0.265	0.432
K-L1 within-dog natural 2-min	130	0.402	0.617
K-L1 within-dog recording	61	0.213	0.355
K-L2 within-dog APP 120-s	1,596	0.343	0.525
K-L2 within-dog natural 2-min	130	0.705	0.892
K-L2 within-dog recording	61	0.292	0.404

(b) Rule 'apply the correction only when the shift visible in CC estimates exceeds τ' ($\hat{\delta}$ rule) and the κ-PACC rule. τ was chosen on S1 windows and frozen; d = AE_CC − AE_rule (positive: rule better); share_corrected: share of samples in which the rule applied the correction.

sample set	rule	tau	AE rule	AE CC	AE always corrected	AE oracle	share corrected	d	ci low	ci high	p	n individuals	p holm	family size	category
S1	PACC	0.255	0.0312	0.0260	0.0527	0.0213	0.103	−0.004	−0.009	+0.000	0.125	12	0.125	1	equivalent
S1	ACC	0.005	0.0324	0.0260	0.0324	0.0204	1.000	−0.006	−0.012	−0.000	0.064	12	0.064	1	inconclusive
S1	kappa-PACC	0.000	0.0527	0.0260	0.0527	0.0213	1.000	−0.024	−0.049	−0.007	0.007	12	0.007	1	worse
S1 natural 2-min	PACC	0.255	0.0451	0.0398	0.0574	0.0312	0.226	−0.007	−0.014	−0.001	0.052	12	0.052	1	inconclusive
S1 natural 2-min	ACC	0.005	0.0392	0.0398	0.0392	0.0295	0.999	−0.001	−0.004	+0.003	0.763	12	0.763	1	equivalent
S1 natural 2-min	kappa-PACC	0.000	0.0574	0.0398	0.0574	0.0312	0.595	−0.018	−0.043	−0.001	0.066	12	0.066	1	inconclusive
S1 APP 120-s	PACC	0.255	0.0985	0.1522	0.0921	0.0816	0.728	+0.054	+0.036	+0.069	0.001	12	0.001	1	better
S1 APP 120-s	ACC	0.005	0.0916	0.1522	0.0916	0.0832	0.998	+0.061	+0.040	+0.079	0.001	12	0.001	1	better
S1 APP 120-s	kappa-PACC	0.000	0.0921	0.1522	0.0921	0.0816	1.000	+0.060	+0.041	+0.077	9.8×10^{-4}	12	9.8×10^{-4}	1	better
S2	PACC	0.255	0.0548	0.0405	0.1155	0.0366	0.118	−0.012	−0.021	−0.004	0.031	12	0.031	1	worse
S2	ACC	0.005	0.0528	0.0405	0.0528	0.0348	1.000	−0.012	−0.019	−0.004	0.017	12	0.017	1	worse
S2	kappa-PACC	0.000	0.1155	0.0405	0.1155	0.0366	1.000	−0.069	−0.124	−0.025	0.004	12	0.004	1	worse
S3 [exploratory]	PACC	0.255	0.1243	0.1442	0.1306	0.1132	0.375	+0.020	−0.004	+0.059	0.500	8	0.500	1	inconclusive
S3 [exploratory]	ACC	0.005	0.1056	0.1442	0.1056	0.1028	1.000	+0.039	+0.003	+0.081	0.070	8	0.070	1	inconclusive
S3 [exploratory]	kappa-PACC	0.000	0.1306	0.1442	0.1306	0.1132	1.000	+0.014	−0.025	+0.062	0.641	8	0.641	1	inconclusive
S4-Smit-closed	PACC	0.255	0.0111	0.0089	0.0111	0.0054	1.000	−0.002	−0.005	+0.002	0.311	12	0.311	1	equivalent
S4-Smit-closed	ACC	0.005	0.0105	0.0089	0.0105	0.0051	1.000	−0.001	−0.004	+0.002	0.459	12	0.459	1	equivalent
S4-Smit-closed	kappa-PACC	0.000	0.0111	0.0089	0.0111	0.0054	0.941	−0.002	−0.005	+0.002	0.311	12	0.311	1	equivalent
S4-Dunford-closed [exploratory]	PACC	0.255	0.0953	0.0869	0.1023	0.0869	0.714	−0.008	−0.016	−0.002	0.125	7	0.125	1	inconclusive
S4-Dunford-closed [exploratory]	ACC	0.005	0.0965	0.0869	0.0965	0.0869	1.000	−0.010	−0.015	−0.003	0.031	7	0.031	1	inconclusive
S4-Dunford-closed [exploratory]	kappa-PACC	0.000	0.1023	0.0869	0.1023	0.0869	0.857	−0.015	−0.023	−0.008	0.031	7	0.031	1	worse

Table S10. Dunford et al. (2024) data: repeated blocks and validation unit.

(a) Seconds per behaviour as reported in Table A2 of Dunford et al. (2024) and recomputed from the published file (rows / 40 Hz): all rows; without the repeated 38–42-row blocks (the version analysed in this study); and, for comparison only, without every exactly duplicated row.

behaviour	TableA2 s	all rows div40 s	block repeats removed s (used here)	every duplicate row removed s (comparison only)
Groom	13.08	13.07	13.07	12.82
Feed	66.98	66.97	66.97	64.60
Rest	2,491.8	2,491.8	2,110.2	1,864.8
Walk	533.65	533.65	508.80	502.18
Trot	86.85	86.85	82.88	81.75
Run	50.80	50.80	49.80	48.30
Shake	7.15	7.15	6.15	5.47
Total	3,250.3	3,250.3	2,837.9	2,579.9

(b) Effect of the validation unit and of the repeated blocks on macro-F1 (our features and pipeline; not a reproduction of the original analysis).

validation	file version	n runs	accuracy	accuracy sd	macroF1	macroF1 sd	macroF1 min	macroF1 max
leave-one-cat-out (pooled predictions)	v2 (repeated blocks removed; used in this study)	1	0.850		0.308		0.308	0.308
leave-one-cat-out (per cat)	v2 (repeated blocks removed; used in this study)	9	0.749	0.218	0.519	0.247	0.300	0.996
random 60/40 split of seconds	v1 (with repeated blocks)	20	0.881	0.011	0.534	0.068	0.419	0.616
random 60/40 split of seconds	v2 (repeated blocks removed; used in this study)	20	0.878	0.009	0.492	0.059	0.435	0.611

Table S11. Seven-class Smit data: bias–variance decomposition of time-budget error by window length and estimator.

ACC and PACC use the confusion matrix estimated on the other cats; ACC-xfit and PACC-xfit use a matrix cross-fitted within the held-out cat (blocks of the opposite type to the window), which is not available in practice. bias_share = bias^2 / MSE.

scenario	window length min	estimator	n cats	n windows	mean minutes	bias2	bias2 lo	bias2 hi	var	var lo	var hi	mse	mse lo	mse hi	AE	AE lo	AE hi	bias share
S1-L2 (harness)	5	CC	12	436	4.439	0.0091	0.0037	0.0165	0.0459	0.0267	0.0675	0.0550	0.0306	0.0834	0.0350	0.0268	0.0440	0.165
S1-L2 (harness)	5	PCC	12	436	4.439	0.0104	0.0057	0.0155	0.0581	0.0425	0.0756	0.0685	0.0484	0.0904	0.0540	0.0458	0.0616	0.152
S1-L2 (harness)	5	ACC	12	436	4.439	0.0136	0.0053	0.0242	0.0646	0.0397	0.0926	0.0782	0.0457	0.1171	0.0393	0.0295	0.0503	0.174
S1-L2 (harness)	5	PACC	12	436	4.439	0.0153	0.0076	0.0248	0.0601	0.0360	0.0879	0.0754	0.0435	0.1127	0.0414	0.0312	0.0526	0.203
S1-L2 (harness)	5	EMQ	12	436	4.439	0.0247	0.0130	0.0405	0.0768	0.0503	0.1017	0.1014	0.0647	0.1405	0.0442	0.0338	0.0547	0.243
S1-L2 (harness)	5	ACC-xfit	12	436	4.439	0.0051	0.0015	0.0097	0.0606	0.0400	0.0858	0.0657	0.0428	0.0918	0.0363	0.0284	0.0452	0.078
S1-L2 (harness)	5	PACC-xfit	12	436	4.439	0.0050	0.0021	0.0098	0.0585	0.0385	0.0836	0.0635	0.0411	0.0895	0.0386	0.0307	0.0477	0.079
S1-L2 (harness)	10	CC	12	220	8.639	0.0094	0.0038	0.0169	0.0325	0.0179	0.0497	0.0420	0.0226	0.0657	0.0329	0.0250	0.0413	0.225
S1-L2 (harness)	10	PCC	12	220	8.639	0.0120	0.0061	0.0188	0.0397	0.0270	0.0541	0.0517	0.0353	0.0691	0.0486	0.0409	0.0560	0.233
S1-L2 (harness)	10	ACC	12	220	8.639	0.0156	0.0065	0.0267	0.0447	0.0256	0.0667	0.0603	0.0321	0.0931	0.0382	0.0283	0.0485	0.259
S1-L2 (harness)	10	PACC	12	220	8.639	0.0184	0.0083	0.0306	0.0476	0.0294	0.0700	0.0660	0.0384	0.1007	0.0429	0.0325	0.0542	0.279
S1-L2 (harness)	10	EMQ	12	220	8.639	0.0332	0.0149	0.0562	0.0610	0.0407	0.0823	0.0942	0.0586	0.1340	0.0473	0.0349	0.0590	0.352
S1-L2 (harness)	10	ACC-xfit	12	220	8.639	0.0050	0.0019	0.0098	0.0441	0.0276	0.0635	0.0491	0.0306	0.0710	0.0355	0.0277	0.0439	0.102
S1-L2 (harness)	10	PACC-xfit	12	220	8.639	0.0057	0.0021	0.0110	0.0497	0.0330	0.0703	0.0554	0.0364	0.0793	0.0404	0.0327	0.0495	0.103
S1-L2 (harness)	30	CC	12	68	24.728	0.0086	0.0036	0.0153	0.0133	0.0053	0.0243	0.0219	0.0093	0.0392	0.0255	0.0194	0.0324	0.392
S1-L2 (harness)	30	PCC	12	68	24.728	0.0121	0.0067	0.0177	0.0170	0.0108	0.0240	0.0292	0.0195	0.0396	0.0390	0.0324	0.0455	0.416
S1-L2 (harness)	30	ACC	12	68	24.728	0.0262	0.0093	0.0531	0.0305	0.0113	0.0560	0.0567	0.0210	0.1076	0.0406	0.0286	0.0549	0.462
S1-L2 (harness)	30	PACC	12	68	24.728	0.0291	0.0121	0.0552	0.0321	0.0173	0.0507	0.0612	0.0309	0.0997	0.0477	0.0350	0.0619	0.476
S1-L2 (harness)	30	EMQ	12	68	24.728	0.0447	0.0140	0.0860	0.0291	0.0180	0.0423	0.0738	0.0338	0.1228	0.0475	0.0321	0.0659	0.606
S1-L2 (harness)	30	ACC-xfit	12	68	24.728	0.0068	0.0021	0.0148	0.0224	0.0125	0.0338	0.0293	0.0155	0.0461	0.0330	0.0246	0.0412	0.233
S1-L2 (harness)	30	PACC-xfit	12	68	24.728	0.0081	0.0027	0.0171	0.0254	0.0156	0.0374	0.0335	0.0196	0.0521	0.0365	0.0279	0.0466	0.241
S1-L2 (harness)	60	CC	10	30	49.302	0.0117	0.0034	0.0233	0.0049	0.0019	0.0090	0.0166	0.0054	0.0320	0.0250	0.0169	0.0352	0.707
S1-L2 (harness)	60	PCC	10	30	49.302	0.0126	0.0051	0.0214	0.0066	0.0047	0.0086	0.0192	0.0112	0.0288	0.0333	0.0266	0.0399	0.654
S1-L2 (harness)	60	ACC	10	30	49.302	0.0407	0.0108	0.0855	0.0166	0.0057	0.0294	0.0572	0.0185	0.1136	0.0465	0.0293	0.0672	0.710
S1-L2 (harness)	60	PACC	10	30	49.302	0.0406	0.0106	0.0880	0.0165	0.0087	0.0248	0.0571	0.0223	0.1100	0.0519	0.0347	0.0721	0.711
S1-L2 (harness)	60	EMQ	10	30	49.302	0.0518	0.0157	0.1130	0.0151	0.0087	0.0211	0.0669	0.0253	0.1300	0.0523	0.0337	0.0743	0.774
S1-L2 (harness)	60	ACC-xfit	10	30	49.302	0.0072	0.0015	0.0167	0.0178	0.0082	0.0303	0.0251	0.0099	0.0468	0.0327	0.0237	0.0427	0.288
S1-L2 (harness)	60	PACC-xfit	10	30	49.302	0.0060	0.0025	0.0104	0.0260	0.0116	0.0450	0.0320	0.0147	0.0531	0.0376	0.0273	0.0483	0.188
S1-L2 (harness)	half recording	CC	12	24	85.544	0.0093	0.0035	0.0175	0.0021	0.0009	0.0036	0.0115	0.0045	0.0207	0.0218	0.0148	0.0301	0.815
S1-L2 (harness)	half recording	PCC	12	24	85.544	0.0117	0.0063	0.0175	0.0023	0.0012	0.0036	0.0139	0.0079	0.0208	0.0282	0.0211	0.0355	0.836
S1-L2 (harness)	half recording	ACC	12	24	85.544	0.0358	0.0128	0.0751	0.0080	0.0014	0.0199	0.0438	0.0146	0.0943	0.0424	0.0296	0.0594	0.817
S1-L2 (harness)	half recording	PACC	12	24	85.544	0.0440	0.0199	0.0815	0.0039	0.0017	0.0069	0.0479	0.0223	0.0878	0.0500	0.0375	0.0656	0.918
S1-L2 (harness)	half recording	EMQ	12	24	85.544	0.0517	0.0200	0.0952	0.0046	0.0021	0.0073	0.0563	0.0237	0.1017	0.0506	0.0343	0.0690	0.919
S1-L2 (harness)	half recording	ACC-xfit	12	24	85.544	0.0045	0.0021	0.0080	0.0119	0.0064	0.0180	0.0164	0.0092	0.0253	0.0303	0.0223	0.0389	0.272
S1-L2 (harness)	half recording	PACC-xfit	12	24	85.544	0.0046	0.0027	0.0069	0.0166	0.0085	0.0251	0.0212	0.0120	0.0319	0.0342	0.0255	0.0436	0.218
S1-L2 (collar)	5	CC	12	436	4.439	0.1357	0.0450	0.2468	0.2119	0.1538	0.2691	0.3476	0.2118	0.4980	0.1026	0.0805	0.1267	0.390
S1-L2 (collar)	5	PCC	12	436	4.439	0.1095	0.0375	0.2011	0.1950	0.1577	0.2364	0.3045	0.2094	0.4089	0.1140	0.0968	0.1318	0.360
S1-L2 (collar)	5	ACC	12	436	4.439	0.1636	0.0557	0.2939	0.2733	0.2043	0.3373	0.4369	0.2692	0.6235	0.1045	0.0786	0.1335	0.374
S1-L2 (collar)	5	PACC	12	436	4.439	0.1571	0.0458	0.2975	0.2544	0.1858	0.3165	0.4115	0.2454	0.5990	0.1067	0.0802	0.1368	0.382
S1-L2 (collar)	5	EMQ	12	436	4.439	0.1512	0.0521	0.2750	0.2870	0.1917	0.3859	0.4381	0.2598	0.6368	0.1107	0.0811	0.1413	0.345
S1-L2 (collar)	5	ACC-xfit	12	436	4.439	0.0589	0.0194	0.1076	0.2927	0.2023	0.3827	0.3517	0.2412	0.4658	0.0978	0.0758	0.1199	0.168
S1-L2 (collar)	5	PACC-xfit	12	436	4.439	0.0343	0.0154	0.0572	0.2328	0.1659	0.3011	0.2671	0.1948	0.3377	0.0883	0.0726	0.1046	0.128
S1-L2 (collar)	10	CC	12	220	8.639	0.1373	0.0439	0.2548	0.1641	0.1161	0.2118	0.3015	0.1717	0.4519	0.0983	0.0755	0.1234	0.456
S1-L2 (collar)	10	PCC	12	220	8.639	0.1120	0.0367	0.2059	0.1491	0.1144	0.1819	0.2610	0.1673	0.3719	0.1053	0.0873	0.1235	0.429
S1-L2 (collar)	10	ACC	12	220	8.639	0.1664	0.0597	0.2954	0.2284	0.1643	0.2913	0.3948	0.2290	0.5840	0.1056	0.0805	0.1341	0.421
S1-L2 (collar)	10	PACC	12	220	8.639	0.1637	0.0482	0.3064	0.2211	0.1586	0.2848	0.3848	0.2162	0.5833	0.1086	0.0819	0.1391	0.425
S1-L2 (collar)	10	EMQ	12	220	8.639	0.1673	0.0581	0.3012	0.2414	0.1516	0.3313	0.4086	0.2187	0.6121	0.1102	0.0777	0.1425	0.409
S1-L2 (collar)	10	ACC-xfit	12	220	8.639	0.0646	0.0192	0.1158	0.2465	0.1682	0.3243	0.3111	0.2026	0.4270	0.0973	0.0744	0.1206	0.208
S1-L2 (collar)	10	PACC-xfit	12	220	8.639	0.0369	0.0170	0.0606	0.1920	0.1334	0.2576	0.2289	0.1596	0.3058	0.0868	0.0703	0.1036	0.161
S1-L2 (collar)	30	CC	12	68	24.728	0.1574	0.0424	0.3070	0.0807	0.0463	0.1177	0.2382	0.1056	0.3941	0.0895	0.0630	0.1194	0.661
S1-L2 (collar)	30	PCC	12	68	24.728	0.1328	0.0368	0.2595	0.0724	0.0479	0.1014	0.2052	0.1047	0.3290	0.0908	0.0703	0.1139	0.647
S1-L2 (collar)	30	ACC	12	68	24.728	0.2257	0.0619	0.4399	0.1338	0.0758	0.1899	0.3595	0.1630	0.5908	0.1093	0.0785	0.1452	0.628
S1-L2 (collar)	30	PACC	12	68	24.728	0.2353	0.0585	0.4739	0.1321	0.0746	0.1867	0.3674	0.1608	0.6340	0.1132	0.0803	0.1519	0.641
S1-L2 (collar)	30	EMQ	12	68	24.728	0.2211	0.0678	0.4135	0.1425	0.0694	0.2204	0.3636	0.1647	0.5867	0.1094	0.0746	0.1452	0.608

continued

scenario	window length min	estimator	n cats	n windows	mean minutes	bias2	bias2 lo	bias2 hi	var	var lo	var hi	mse	mse lo	mse hi	AE	AE lo	AE hi	bias share
S1-L2 (collar)	30	ACC-xfit	12	68	24.728	0.0543	0.0221	0.0939	0.1675	0.0950	0.2439	0.2218	0.1257	0.3213	0.0856	0.0624	0.1114	0.245
S1-L2 (collar)	30	PACC-xfit	12	68	24.728	0.0525	0.0293	0.0761	0.1439	0.0750	0.2144	0.1963	0.1139	0.2833	0.0820	0.0614	0.1044	0.267
S1-L2 (collar)	60	CC	10	30	49.302	0.1081	0.0328	0.2104	0.0499	0.0191	0.0833	0.1580	0.0524	0.2935	0.0737	0.0489	0.0994	0.684
S1-L2 (collar)	60	PCC	10	30	49.302	0.0902	0.0332	0.1836	0.0452	0.0242	0.0709	0.1354	0.0629	0.2434	0.0762	0.0568	0.0976	0.666
S1-L2 (collar)	60	ACC	10	30	49.302	0.1664	0.0720	0.2957	0.1079	0.0591	0.1633	0.2743	0.1401	0.4430	0.1058	0.0846	0.1278	0.607
S1-L2 (collar)	60	PACC	10	30	49.302	0.1545	0.0675	0.2971	0.0939	0.0468	0.1427	0.2484	0.1299	0.4220	0.1097	0.0872	0.1328	0.622
S1-L2 (collar)	60	EMQ	10	30	49.302	0.1660	0.0605	0.3205	0.0986	0.0315	0.1857	0.2646	0.1002	0.4560	0.0995	0.0677	0.1294	0.627
S1-L2 (collar)	60	ACC-xfit	10	30	49.302	0.0641	0.0133	0.1565	0.0867	0.0449	0.1381	0.1508	0.0651	0.2843	0.0760	0.0560	0.1006	0.425
S1-L2 (collar)	60	PACC-xfit	10	30	49.302	0.0401	0.0085	0.0983	0.0848	0.0483	0.1218	0.1249	0.0644	0.2016	0.0747	0.0563	0.0938	0.321
S1-L2 (collar)	half recording	CC	12	24	85.544	0.1368	0.0466	0.2459	0.0095	0.0044	0.0160	0.1463	0.0539	0.2526	0.0785	0.0528	0.1062	0.935
S1-L2 (collar)	half recording	PCC	12	24	85.544	0.1118	0.0398	0.2034	0.0088	0.0042	0.0145	0.1206	0.0478	0.2101	0.0715	0.0522	0.0954	0.927
S1-L2 (collar)	half recording	ACC	12	24	85.544	0.2874	0.1026	0.5264	0.0274	0.0093	0.0519	0.3148	0.1315	0.5462	0.1194	0.0851	0.1568	0.913
S1-L2 (collar)	half recording	PACC	12	24	85.544	0.2810	0.0992	0.5324	0.0307	0.0088	0.0604	0.3117	0.1384	0.5522	0.1233	0.0935	0.1570	0.901
S1-L2 (collar)	half recording	EMQ	12	24	85.544	0.2291	0.0817	0.4241	0.0386	0.0087	0.0826	0.2677	0.1066	0.4642	0.1067	0.0685	0.1455	0.856
S1-L2 (collar)	half recording	ACC-xfit	12	24	85.544	0.0416	0.0111	0.0923	0.0859	0.0381	0.1540	0.1275	0.0611	0.2104	0.0793	0.0560	0.1038	0.326
S1-L2 (collar)	half recording	PACC-xfit	12	24	85.544	0.0665	0.0147	0.1333	0.0794	0.0339	0.1426	0.1459	0.0705	0.2443	0.0806	0.0595	0.1059	0.456
S2-L2 (mapped)	5	CC	12	436	4.439	0.3295	0.1522	0.5663	0.4827	0.3748	0.5748	0.8122	0.6113	1.0402	0.1691	0.1434	0.1949	0.406
S2-L2 (mapped)	5	PCC	12	436	4.439	0.2530	0.1353	0.4061	0.4044	0.3207	0.4817	0.6573	0.5279	0.8079	0.1805	0.1650	0.1978	0.385
S2-L2 (mapped)	5	ACC	12	436	4.439	0.3436	0.1626	0.5823	0.5343	0.4189	0.6294	0.8778	0.6686	1.1042	0.1746	0.1483	0.2004	0.391
S2-L2 (mapped)	5	PACC	12	436	4.439	0.3309	0.1790	0.5375	0.5122	0.4030	0.6076	0.8430	0.6618	1.0466	0.1897	0.1697	0.2110	0.392
S2-L2 (mapped)	5	EMQ	12	436	4.439	0.4275	0.2728	0.6435	0.7114	0.5765	0.8306	1.1389	0.9841	1.3037	0.2188	0.1979	0.2371	0.375
S2-L2 (mapped)	5	ACC-xfit	12	436	4.439	0.0719	0.0441	0.1006	0.3558	0.2818	0.4239	0.4277	0.3336	0.5116	0.1149	0.0938	0.1327	0.168
S2-L2 (mapped)	5	PACC-xfit	12	436	4.439	0.0267	0.0145	0.0416	0.2504	0.1964	0.3017	0.2772	0.2169	0.3321	0.0947	0.0799	0.1066	0.096
S2-L2 (mapped)	10	CC	12	220	8.639	0.3312	0.1524	0.5624	0.3853	0.2972	0.4684	0.7165	0.5101	0.9523	0.1614	0.1363	0.1877	0.462
S2-L2 (mapped)	10	PCC	12	220	8.639	0.2578	0.1416	0.4107	0.3227	0.2473	0.3933	0.5805	0.4453	0.7390	0.1689	0.1522	0.1885	0.444
S2-L2 (mapped)	10	ACC	12	220	8.639	0.3539	0.1712	0.5894	0.4279	0.3386	0.5120	0.7819	0.5688	1.0167	0.1695	0.1448	0.1953	0.453
S2-L2 (mapped)	10	PACC	12	220	8.639	0.3442	0.1914	0.5448	0.4217	0.3211	0.5079	0.7659	0.5888	0.9748	0.1843	0.1658	0.2054	0.449
S2-L2 (mapped)	10	EMQ	12	220	8.639	0.4519	0.2979	0.6504	0.6018	0.4825	0.7060	1.0537	0.9081	1.2233	0.2151	0.1980	0.2326	0.429
S2-L2 (mapped)	10	ACC-xfit	12	220	8.639	0.0680	0.0427	0.0926	0.3005	0.2261	0.3701	0.3685	0.2817	0.4457	0.1124	0.0913	0.1310	0.185
S2-L2 (mapped)	10	PACC-xfit	12	220	8.639	0.0300	0.0143	0.0515	0.2048	0.1570	0.2508	0.2348	0.1796	0.2870	0.0936	0.0786	0.1076	0.128
S2-L2 (mapped)	30	CC	12	68	24.728	0.3718	0.1643	0.6712	0.2291	0.1432	0.3229	0.6009	0.4048	0.8703	0.1500	0.1202	0.1825	0.619
S2-L2 (mapped)	30	PCC	12	68	24.728	0.2948	0.1573	0.4893	0.1846	0.1104	0.2662	0.4794	0.3523	0.6509	0.1532	0.1341	0.1770	0.615
S2-L2 (mapped)	30	ACC	12	68	24.728	0.4090	0.1922	0.7208	0.2644	0.1686	0.3627	0.6734	0.4642	0.9458	0.1627	0.1316	0.1941	0.607
S2-L2 (mapped)	30	PACC	12	68	24.728	0.4089	0.2179	0.6793	0.2516	0.1587	0.3532	0.6606	0.4795	0.9021	0.1747	0.1516	0.2012	0.619
S2-L2 (mapped)	30	EMQ	12	68	24.728	0.5356	0.3398	0.8027	0.3742	0.2539	0.4911	0.9098	0.7333	1.1307	0.2084	0.1841	0.2327	0.589
S2-L2 (mapped)	30	ACC-xfit	12	68	24.728	0.0744	0.0413	0.1154	0.1584	0.1008	0.2239	0.2329	0.1480	0.3253	0.0964	0.0736	0.1199	0.320
S2-L2 (mapped)	30	PACC-xfit	12	68	24.728	0.0439	0.0222	0.0703	0.1108	0.0730	0.1514	0.1546	0.1040	0.2061	0.0820	0.0649	0.0987	0.284
S2-L2 (mapped)	60	CC	10	30	49.302	0.3856	0.1929	0.6505	0.1532	0.0791	0.2336	0.5388	0.3563	0.7915	0.1560	0.1297	0.1863	0.716
S2-L2 (mapped)	60	PCC	10	30	49.302	0.2981	0.1771	0.4629	0.1198	0.0667	0.1717	0.4178	0.3051	0.5763	0.1477	0.1298	0.1704	0.713
S2-L2 (mapped)	60	ACC	10	30	49.302	0.4422	0.2286	0.7305	0.1838	0.0990	0.2783	0.6261	0.4300	0.8872	0.1697	0.1449	0.1974	0.706
S2-L2 (mapped)	60	PACC	10	30	49.302	0.4397	0.2717	0.6652	0.1619	0.0918	0.2321	0.6016	0.4458	0.8174	0.1765	0.1590	0.1980	0.731
S2-L2 (mapped)	60	EMQ	10	30	49.302	0.6060	0.4264	0.8260	0.2173	0.1277	0.3107	0.8233	0.6369	1.0323	0.2133	0.1892	0.2361	0.736
S2-L2 (mapped)	60	ACC-xfit	10	30	49.302	0.0613	0.0276	0.0988	0.0872	0.0540	0.1235	0.1485	0.0859	0.2207	0.0868	0.0663	0.1112	0.413
S2-L2 (mapped)	60	PACC-xfit	10	30	49.302	0.0540	0.0276	0.0855	0.0718	0.0402	0.1067	0.1258	0.0674	0.1903	0.0808	0.0609	0.1014	0.429
S2-L2 (mapped)	half recording	CC	12	24	85.544	0.3416	0.1595	0.5792	0.0208	0.0079	0.0381	0.3624	0.1834	0.5995	0.1282	0.0976	0.1626	0.943
S2-L2 (mapped)	half recording	PCC	12	24	85.544	0.2650	0.1465	0.4226	0.0185	0.0073	0.0316	0.2835	0.1652	0.4371	0.1248	0.1027	0.1515	0.935
S2-L2 (mapped)	half recording	ACC	12	24	85.544	0.4085	0.2010	0.6710	0.0260	0.0093	0.0503	0.4345	0.2340	0.6977	0.1445	0.1144	0.1783	0.940
S2-L2 (mapped)	half recording	PACC	12	24	85.544	0.3910	0.2265	0.6129	0.0265	0.0091	0.0485	0.4174	0.2545	0.6243	0.1498	0.1265	0.1773	0.937
S2-L2 (mapped)	half recording	EMQ	12	24	85.544	0.6005	0.4105	0.8238	0.0573	0.0243	0.1061	0.6579	0.4821	0.8641	0.1922	0.1656	0.2156	0.913
S2-L2 (mapped)	half recording	ACC-xfit	12	24	85.544	0.1275	0.0630	0.2061	0.1166	0.0713	0.1690	0.2441	0.1637	0.3280	0.1167	0.0939	0.1400	0.522
S2-L2 (mapped)	half recording	PACC-xfit	12	24	85.544	0.0315	0.0119	0.0641	0.0815	0.0431	0.1203	0.1129	0.0569	0.1781	0.0841	0.0617	0.1082	0.279